MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GENERATOR, FUNCTION, DIGITAL, KEY CODE UHX

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic function generator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.876 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be UHX.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/194-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

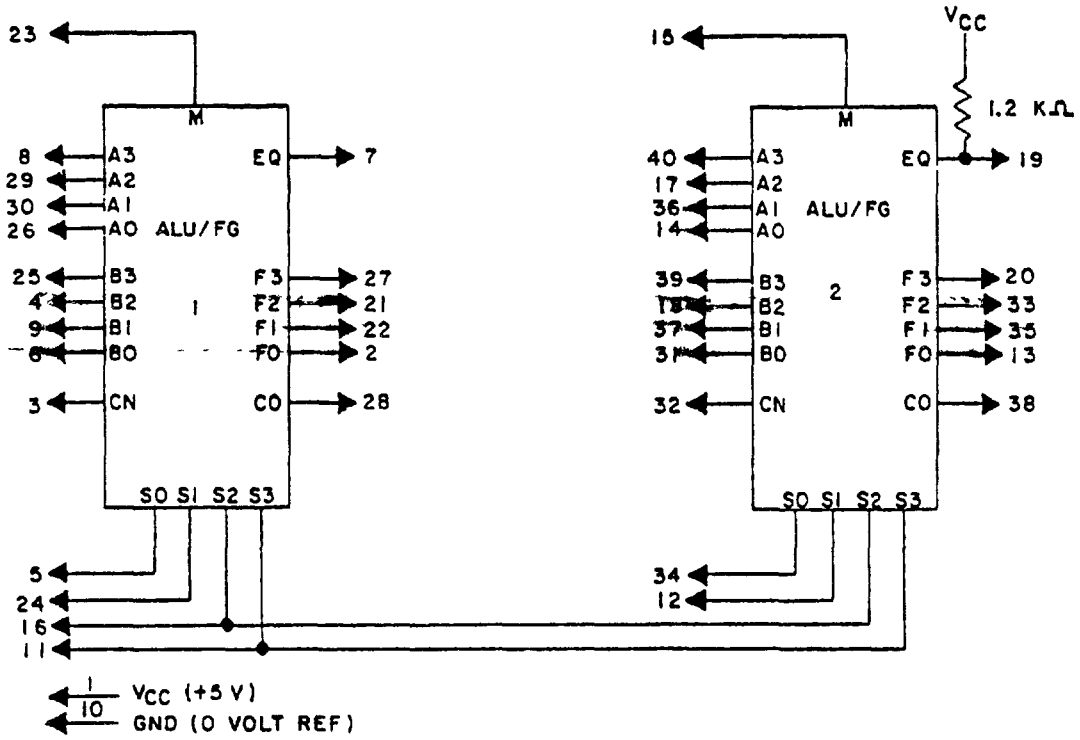
3.9 Functional test. All modules shall be subjected to a functional test which conforms to the requirements of 3.9.1 through 3.9.3.

3.9.1 Test patterns. A set of unique test patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables I and XXIII satisfy or exceed these unique test patterns, no additional tests shall be added; otherwise, the needed tests shall be added to table XXIII. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at H state" or a "stuck at L state", the vendor shall be required to submit to the contracting activity and the Standard Electronic Modules Program Quality Assurance Activity the proposed new set of test patterns for approval.

3.9.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1%.
High level - Input voltage shall be 2.0 V ±1%.

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NOTES:
1. A0 - A3 = Word "A" inputs.
2. B0 - B3 = Word "B" inputs.
3. S0 - S3 = Function select inputs.
4. CN = Carry input.
5. M = Mode control input.
6. F0 - F3 = Function outputs.
7. EQ = Comparator output.
8. CO = Carry output.
9. EQ output pin 7 is an open collector output (see 6.5.2).
10. EQ output pin 19 is an open collector output pulled up to V_{CC} through a 1.2 kΩ resistor (see 1.3.2).

FIGURE 1. UHX module function diagram, function generator.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature -0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	IV	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B12	V_{OUT} (low loaded)	V	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C12, C15-C26	Input breakdown current	VI	S	0	1	0	1	0	1	0	1	mA
C13,C14	Input breakdown current	VI	S	0	2	0	2	0	2	0	2	mA
D1-D8, D11,D12	Output short circuit current	VII	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
D9	Output short circuit current	VII	X	-3.5	-5.5	-3.5	-5.5	-3.5	-5.5	-3.5	-5.5	mA
D10	Output short circuit current	VII	X	-0.0	-0.5	-0.0	-0.5	-0.0	-0.5	-0.0	-0.5	mA
E1-E12	Output leakage current	VIII	X	0	1	0	1	0	1	0	1	mA
F1-F26	Input clamp diode voltage	IX	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G12	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
H1-H12	Output voltage, low level	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
J1,J2	Input current, low level	XII	X	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	mA
J3-J18	Input current, low level	XII	X	-1.2	-6.6	-1.2	-6.6	-1.2	-6.6	-1.2	-6.6	mA
J19-J22	Input current, low level	XII	X	-1.6	-8.8	-1.6	-8.8	-1.6	-8.8	-1.6	-8.8	mA
J23,J24	Input current, low level	XII	X	-3.2	-17.6	-3.2	-17.6	-3.2	-17.6	-3.2	-17.6	mA
J25,J26	Input current, low level	XII	X	-2	-11	-2	-11	-2	-11	-2	-11	mA
K1,K2	Input current, high level	XIII	X	0	50	0	50	0	50	0	50	µA
K3-K18	Input current, high level	XIII	X	0	150	0	150	0	150	0	150	µA
K19-K22	Input current, high level	XIII	X	0	200	0	200	0	200	0	200	µA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K23,K24	Input current, high level	XIII	X	0	400	0	400	0	400	0	400	µA
K25,K26	Input current, high level	XIII	X	0	250	0	250	0	250	0	250	µA
L1,L2	Power supply current, high level and low level	XIV	S	1	440	1	440	1	440	1	440	mA
M1-M10	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2	15	2	15	2	18	2	18	ns
M11-M20	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2	30	2	30	2	30	2	30	ns
M21,M22	Transition time (t_{TLH}), C_L = 50 pF	XV	S	10	40	10	40	10	45	10	45	ns
M23,M24	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	10	65	10	65	10	70	10	70	ns
N1-N10, N21,N22	Transition time (t_{THL}), C_L = 50 pF	XV	S	1	10	1	10	1	10	1	10	ns
N11-N20, N23,N24	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2	20	2	20	2	20	2	20	ns
P1-P8, P17-P24	Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	XVI	S	5	21	5	21	5	24	5	24	ns
P9-P16, P25-P32	Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	XVI	Q	8	24	8	24	8	27	8	27	ns
P33-P40	Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	XVI	S	3	17	3	17	3	19	3	19	ns
P41-P48	Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	XVI	Q	5	20	5	20	5	22	5	22	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P49,P50	Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	XVI	S	3	12	3	12	3	14	3	14	ns
P51,P52	Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	XVI	Q	5	15	5	15	5	16	5	16	ns
P53-P60, P69-P76	Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	XVI	S	5	20	5	20	5	23	5	23	ns
P61-P68, P77-P84	Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	XVI	Q	8	23	8	23	8	25	8	25	ns
P85-P92, P101-P108	Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	XVI	S	5	24	5	24	5	26	5	26	ns
P93-P100, P109-P116	Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	XVI	Q	8	27	8	27	8	29	8	29	ns
P117-P124	Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	XVI	S	5	17	5	17	5	19	5	19	ns
P125-P132	Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	XVI	Q	8	20	8	20	8	22	8	22	ns
P133-P140, P149-P156	Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	XVI	S	5	27	5	27	5	29	5	29	ns
P141-P148, P157-P164	Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	XVI	Q	8	31	8	31	8	33	8	33	ns
P165-P172, P181-P188	Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	XVI	S	10	42	10	42	10	44	10	44	ns
P173-P180, P189, P193-P196	Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	XVI	Q	10	45	10	45	10	47	10	47	ns
P190-P192	Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	XVI	Q	10	50	10	50	10	52	10	52	ns
P197-P204, P213-P220	Propagation delay time (t_{PLH}), logic mode, C_L = 50 pF	XVI	S	5	24	5	24	5	26	5	26	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P205-P212, P221-P228	Propagation delay time (t_{PLH}), logic mode, C_L = 150 pF	XVI	Q	8	28	8	28	8	30	8	30	ns
Q1-Q8, Q17-Q24	Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	XVII	S	5	21	5	21	5	23	5	23	ns
Q9-Q16, Q25-Q32	Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	XVII	Q	8	24	8	24	8	26	8	26	ns
Q33-Q40	Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	XVII	S	3	17	3	17	3	19	3	19	ns
Q41-Q48	Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	XVII	Q	5	20	5	20	5	22	5	22	ns
Q49,Q50	Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	XVII	S	3	14	3	14	3	16	3	16	ns
Q51,Q52	Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	XVII	Q	5	18	5	18	5	20	5	20	ns
Q53-Q60, Q69-Q76	Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	XVII	S	5	20	5	20	5	22	5	22	ns
Q61-Q68, Q77-Q84	Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	XVII	Q	8	24	8	24	8	26	8	26	ns
Q85-Q92, Q101-Q108	Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	XVII	S	5	26	5	26	5	28	5	28	ns
Q93-Q100, Q109-Q116	Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	XVII	Q	8	30	8	30	8	32	8	32	ns
Q117-Q124	Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	XVII	S	3	17	3	17	3	19	3	19	ns
Q125-Q132	Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	XVII	Q	5	20	5	20	5	22	5	22	ns
Q133-Q140, Q149-Q156	Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	XVII	S	5	27	5	27	5	29	5	29	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q141-Q148, Q157-Q164	Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	XVII	Q	- 8	31	8	31	8	33	8	33	ns
Q165-Q172, Q181-Q188	Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	XVII	S	10	33	10	33	10	35	10	35	ns
Q173-Q180, Q189-Q196	Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	XVII	Q	10	37	10	37	10	40	10	40	ns
Q197-Q204, Q213-Q220	Propagation delay time (t_{PHL}), logic mode, C_L = 50 pF	XVII	S	5	26	5	26	5	28	5	28	ns
Q205-Q212, Q221-Q228	Propagation delay time (t_{PHL}), logic mode, C_L = 150 pF	XVII	Q	8	30	8	30	8	32	8	32	ns
R1-R12	Output crosstalk, low level	XVIII	Q	0	900	0	900	0	900	0	900	mV
R13-R24	Output crosstalk, delayed, low level	XVIII	Q	0	500	0	500	0	500	0	500	mV
S1-S12	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XXI	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
V1-V24	Input capacitance	XXII	Q	2	25	2	25	2	25	2	25	pF
V25,V26	Input capacitance, S2, S3	XXII	Q	2	40	2	40	2	40	2	40	pF
W1-W23	Functional test	XXIII	X	← See table XXIII →								

1/ "X" parameters shall be 100 percent tested at 25°C during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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3.9.3 Output requirements. The output requirements for the test patterns shall be as follows:

Low level - Output voltage shall be 0 V minimum, 0.5 V maximum.
High level - Output voltage shall be 2.5 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

*4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4 4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U and W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U and W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/194-1 supersedes part number M28787/194. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation		2.42 W (outputs not loaded)
Input current		
Mode control		
Low level	(V_{IL} = 0.0 V)	-2.2 mA maximum
	(V_{IL} = 0.5 V)	-2.0 mA maximum
High level	(V_{IH} = 2.5 V)	50 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
A or B date		
Low level	(V_{IL} = 0.0 V)	-6.6 mA maximum
	(V_{IL} = 0.5 V)	-6.0 mA maximum
High level	(V_{IH} = 2.5 V)	150 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Independent select line		
Low level	(V_{IL} = 0.0 V)	-8.8 mA maximum
	(V_{IL} = 0.5 V)	-8.0 mA maximum
High level	(V_{IH} = 2.5 V)	200 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Common select line		
Low level	(V_{IL} = 0.0 V)	-17.6 mA maximum
	(V_{IL} = 0.5 V)	-16.0 mA maximum
High level	(V_{IH} = 2.5 V)	400 µA maximum
	(V_{IH} = 5.5 V)	2.0 mA maximum
Carry		
Low level	(V_{IL} = 0.0 V)	-11.0 mA maximum
	(V_{IL} = 0.5 V)	-10.0 mA maximum
High level	(V_{IH} = 2.5 V)	250 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Input threshold voltage		
Low level		0.8 V maximum
High level		2.0 V minimum
Output voltage		
Low level		0.5 V maximum
High level		2.5 V minimum
Output drive current		
Low level	(V_{OL} = 0.5 V)	
All outputs except 2EQ		20 mA maximum
2EQ output		16 mA maximum
High level		
All outputs except 1EQ		
(V_{OH} = 2.5 V)		-1 mA minimum
1EQ output	(V_{OH} = 5.5 V)	250 µA minimum

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6.5 Module function. This module contains two arithmetic logic unit function generators (ALU/FG).

6.5.1 Input to output relationship. Each ALU/FG has the input to output relationship of table II. Each ALU/FG will perform either logic functions (M = High level) or arithmetic functions (M = Low level). The function performed on the A and B inputs is determined by the logic state of the select inputs (S0, S1, S2, and S3). Logic functions are performed on a bit basis. All arithmetic functions are performed on a word basis where each word consists of four bits. The A word is composed of bits A0, A1, A2, and A3, where A0 is the least significant bit. The B word is composed of bits B0, B1, B2, and B3 where B0 is considered the least significant bit. The carry input (CN) and the ripple carry output (C0) are provided for arithmetic operations. When the ALU/FG is operated in the logic mode (M = H), the C0 output is considered a "don't care condition".

TABLE II. Input to output relationship. 1/, 2/

Selection				Logic functions M = H	Arithmetic operations M = L	
S3	S2	S1	S0		CN = H	CN = L
L	L	L	L	$F = \overline{A}$	F = A	F = A PLUS 1
L	L	L	H	$F = \overline{A + B}$	F = A + B	F = (A + B) PLUS 1
L	L	H	L	$F = \overline{A} \cdot B$	$F = A + \overline{B}$	$F = (A + \overline{B})$ PLUS 1
L	L	H	H	F = 0	F = MINUS 1 (2'S Complement)	F = Zero
L	H	L	L	$F = \overline{A \cdot B}$	$F = A$ PLUS $(A \cdot \overline{B})$	$F = A$ PLUS $(A \cdot \overline{B})$ PLUS 1
L	H	L	H	$F = \overline{B}$	$F = (A + B)$ PLUS $(A \cdot \overline{B})$	$F = (A + B)$ PLUS $(A \cdot \overline{B})$ PLUS 1
L	H	H	L	$F = A \oplus B$	F = A MINUS B MINUS 1	F = A MINUS B
L	H	H	H	$F = A \cdot \overline{B}$	$F = (A \cdot \overline{B})$ MINUS 1	$F = A \cdot \overline{B}$
H	L	L	L	$F = \overline{A} + B$	F = A PLUS (A · B)	F = A PLUS (A · B) PLUS 1
H	L	L	H	$F = \overline{A \oplus B}$	F = A PLUS B	F = A PLUS B PLUS 1
H	L	H	L	F = B	$F = (A + \overline{B})$ PLUS (A · B)	$F = (A + \overline{B})$ PLUS (A · B) PLUS 1
H	L	H	H	F = A . B	F = (A · B) MINUS 1	F = A · B
H	H	L	L	F = 1	F = A PLUS A (See note 2)	F = A PLUS A PLUS 1
H	H	L	H	$F = A + \overline{B}$	F = (A + B) PLUS A	F = (A + B) PLUS A PLUS 1
H	H	H	L	F = A + B	$F = (A + \overline{B})$ PLUS A	$F = (A + \overline{B})$ PLUS A PLUS 1
H	H	H	H	F = A	F = A MINUS 1	F = A

1/ L = Low-level voltage.
H = High-level voltage.
$\overline{}$ = Complement operator.
+ = OR operator.
· = AND operator.
$\oplus$ = Exclusive OR operator.

2/ Each bit is shifted to the next more significant position.

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Placing the ALU/FG in the subtract mode (S0 = L, S1 = H, S2 = H, and S3 = L) produces a CO output that represents A and B word magnitude comparisons. The truth table for magnitude comparison is shown in table III. When two words of equal magnitude are applied to the A and B inputs with the ALU/FG in the subtract mode and CN = H, the EQ output will go to a high level. For all other cases with the ALU/FG in the subtract mode and CN = H, the EQ output will be a low level. The EQ output will be a high level in all other cases where the function outputs (F0, F1, F2, and F3) all are high level.

TABLE III. Magnitude comparison outputs. 1/, 2/

Word comparisons	Input CN	Output CO
A $\leq$ B	H	H
A < B	L	H
A > B	H	L
A $\geq$ B	L	L

1/ H = High-level voltage.
2/ L = Low-level voltage.

6.5.2. Parallel outputs. Paralleling of outputs is not permissible except for the EQ outputs. To allow comparison of more than two words paralleling of EQ outputs is permitted providing the following constraints are observed:

EQ output pin 7 shall be an open collector while EQ output pin 19 shall be connected to a 1.2 kΩ resistor tied to the output and returned to V_{CC}. When using only the EQ output from pin 7, a pull-up resistor equal to any value from 750Ω to 1.8 kΩ must be connected from pin 7 to V_{CC}. When paralleling outputs, a maximum of three pin 7 outputs may be tied to each pin 19 output. No more than four of any EQ outputs may be tied together.

6.6 Test circuit decoupling. Decoupling of power supplies and test circuits to reduce noise is permissible for all tests except test condition T, power supply transient amplitude.

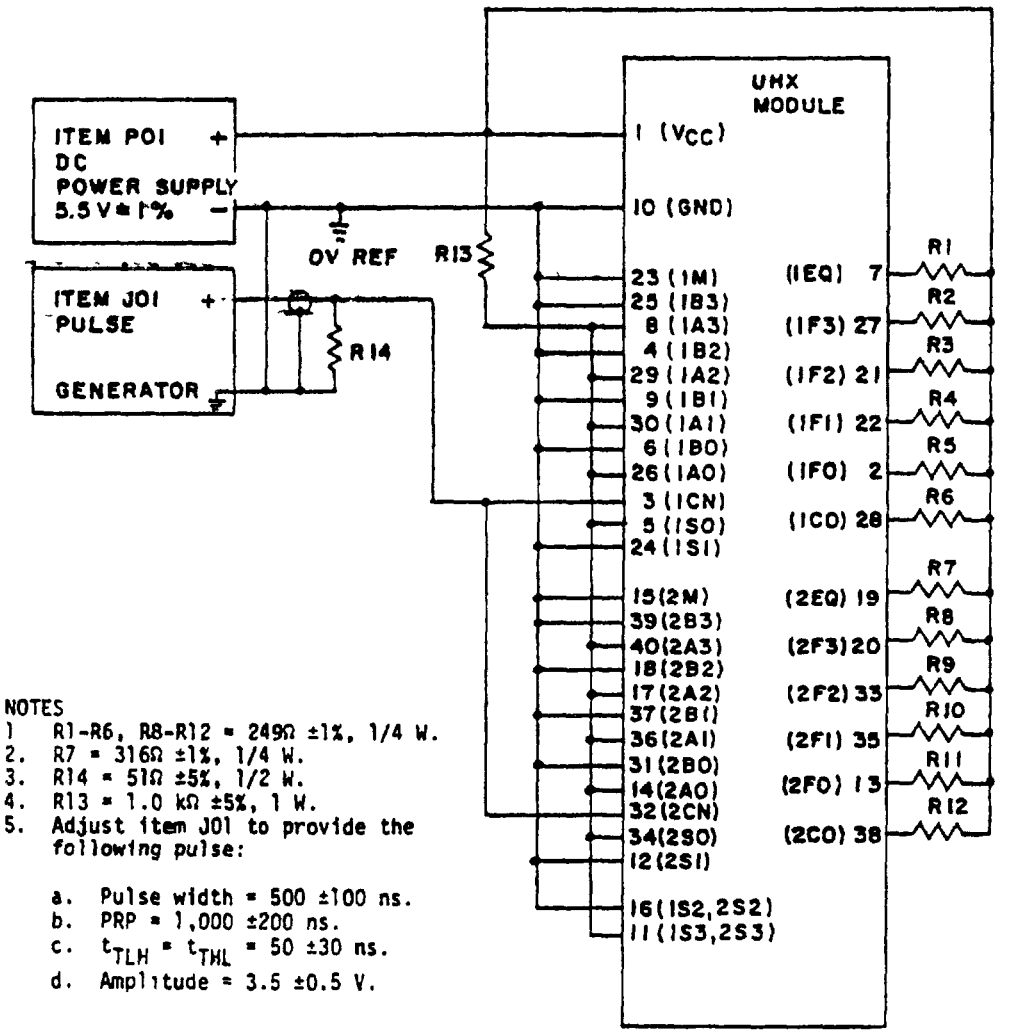
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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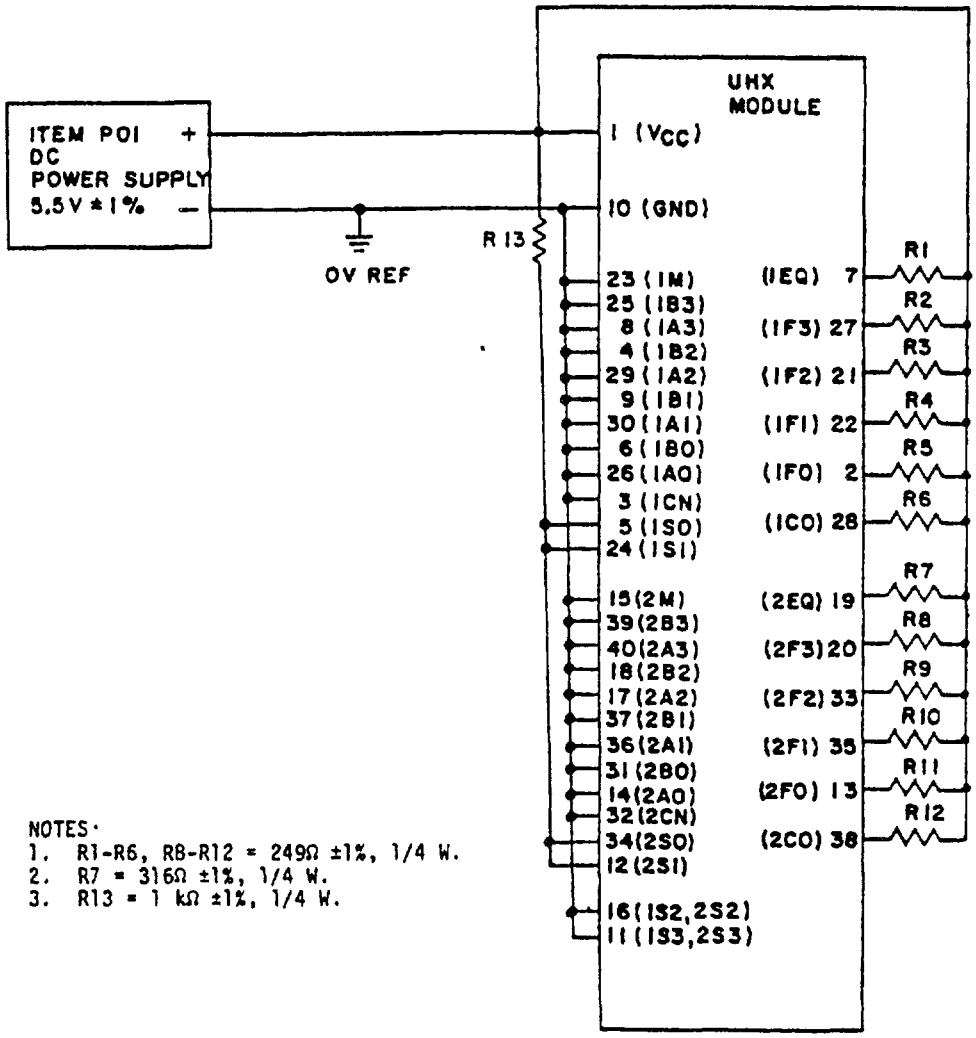


NOTES
1. R1-R6, R8-R12 = 249Ω ±1%, 1/4 W.
2. R7 = 316Ω ±1%, 1/4 W.
3. R14 = 51Ω ±5%, 1/2 W.
4. R13 = 1.0 kΩ ±5%, 1 W.
5. Adjust item J01 to provide the following pulse:

   a. Pulse width = 500 ±100 ns.
   b. PRP = 1,000 ±200 ns.
   c. $t_{TLH} = t_{THL}$ = 50 ±30 ns.
   d. Amplitude = 3.5 ±0.5 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R6, R8-R12 = 249Ω ±1%, 1/4 W.
2. R7 = 316Ω ±1%, 1/4 W.
3. R13 = 1 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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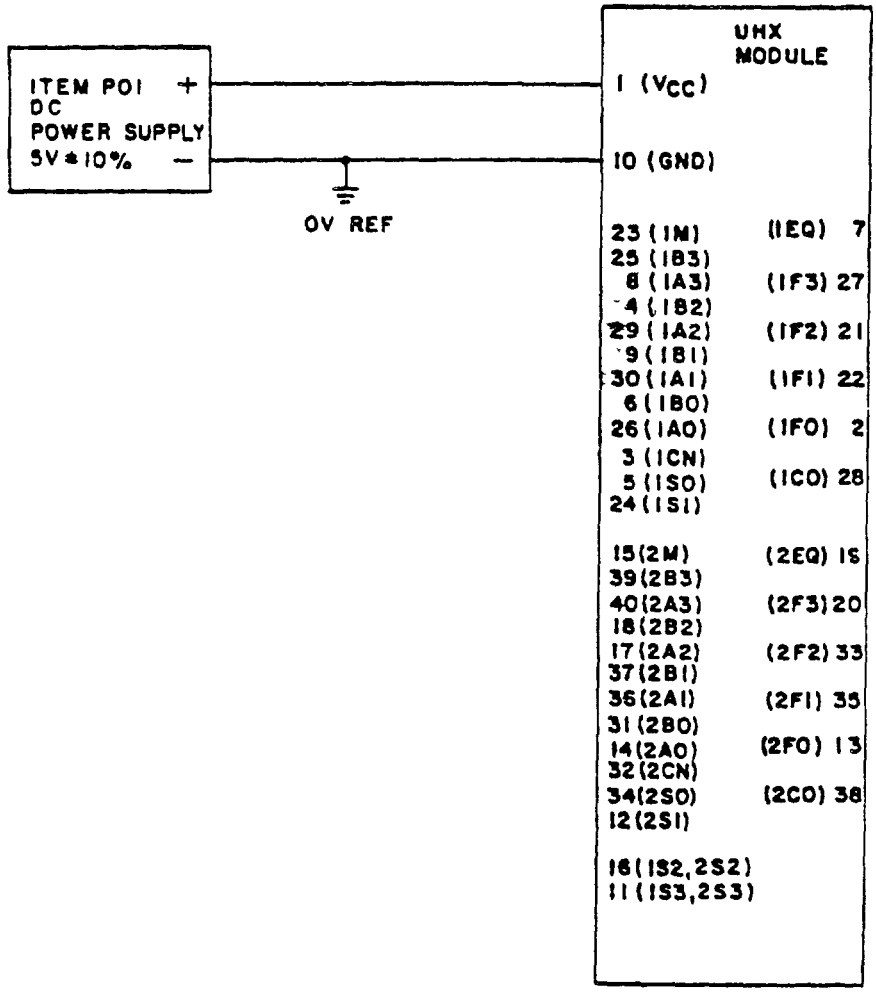


FIGURE 4. Vibration test circuit.

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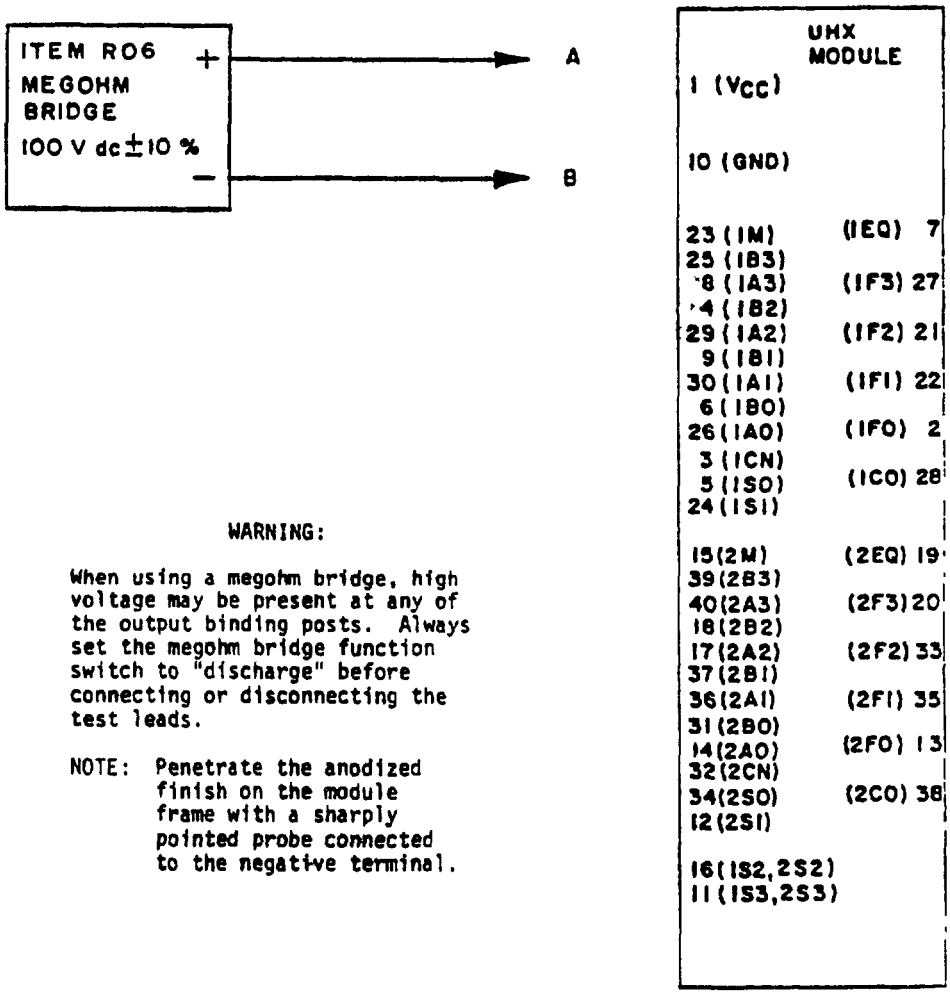
TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1F3 V_{OUT} (low loaded)	27	Open	0.01	1.00	V	6
B2	1F2 "	21	"	"	"	"	"
B3	1F1 "	22	"	"	"	"	"
B4	1F0 "	2	"	"	"	"	"
B5	1C0 "	28	"	"	"	"	"
B6	1EQ "	7	"	"	"	"	"
B7	2F3 "	38	"	"	"	"	"
B8	2F2 "	20	"	"	"	"	"
B9	2F1 "	33	"	"	"	"	"
B10	2F0 "	35	"	"	"	"	"
B11	2C0 "	13	"	"	"	"	"
B12	2EQ "	Open	19	"	"	"	"

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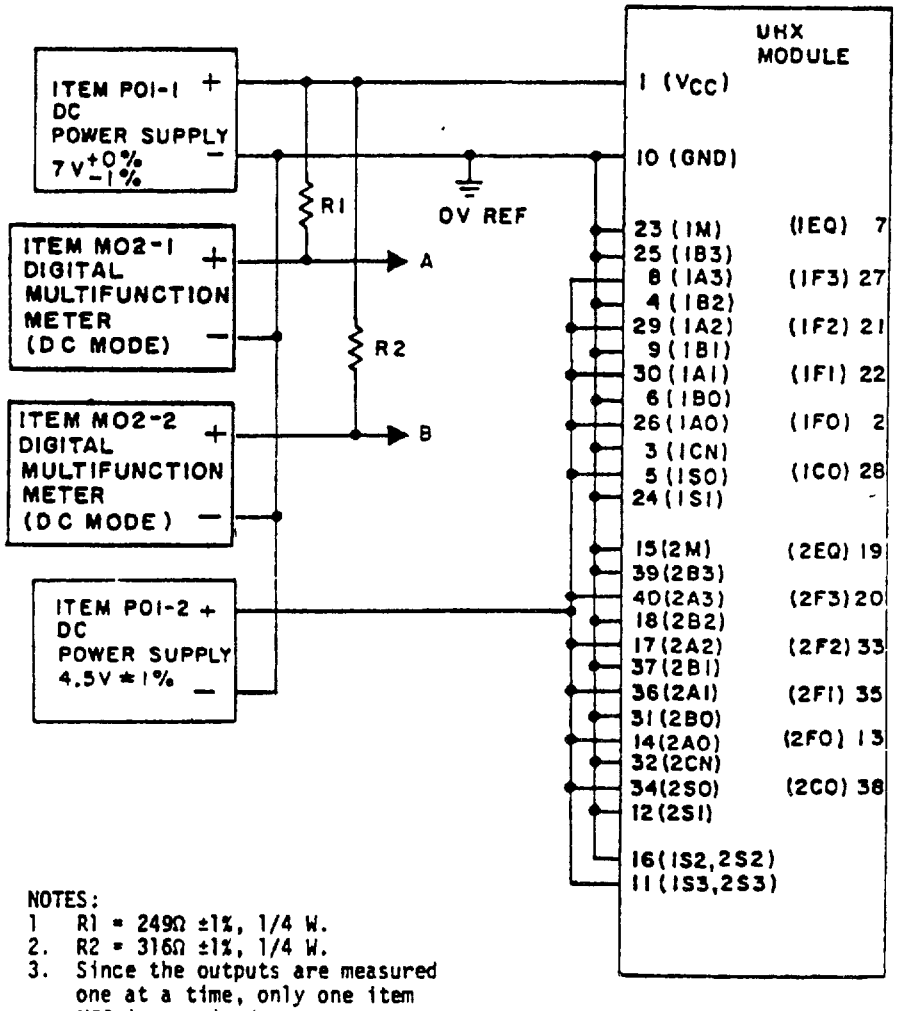
WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table IV.

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NOTES:
1. R1 = 249Ω ±1%, 1/4 W.
2. R2 = 316Ω ±1%, 1/4 W.
3. Since the outputs are measured one at a time, only one item MO2 is required.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1B3 Input breakdown current	25	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,26,29,30, 31,32,34,36, 37,39,40	0	1	mA	7
C2	1A3 "	8	3,4,5,6,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C3	1B2 "	4	3,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C4	1A2 "	29	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,30, 31,32,34,36, 37,39,40	"	"	"	"
C5	1B1 "	9	3,4,5,6,8, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C6	1A1 "	30	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 31,32,34, 36,37,39,40	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C7	1BO Input breakdown current	6	3,4,5,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	0	1	mA	7
C8	1AO "	26	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,29,30, 31,32,34,36, 37,39,40	"	"	"	"
C9	1CN "	3	4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C10	1M "	23	3,4,5,6,8,9, 11,12,14,15, 16,17,18, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C11	1SO "	5	3,4,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C12	1S1 "	24	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C13	1S2, 2S2 Input breakdown current	16	3,4,5,6,8,9, 11,12,14,15, 17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	0	2	mA	7
C14	1S3, 2S3 "	11	3,4,5,6,8,9, 12,14,15,16, 17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	2	"	"
C15	2S0 "	34	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,36, 37,39,40	"	1	"	"
C16	2S1 "	12	3,4,5,6,8,9, 11,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C17	2N "	15	3,4,5,6,8,9, 11,12,14,16, 17,18,23,24, 25,26,29,30, 31,32,34,36, 37,39,40	"	"	"	"
C18	2CN "	32	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,34,36, 37,39,40	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	2B3 Input breakdown current	39	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,40	0	1	mA	7
C20	2A3 "	40	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39	"	"	"	"
C21	2B2 "	18	3,4,5,6,8,9, 11,12,14,15, 16,17,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C22	2A2 "	17	3,4,5,6,8,9, 11,12,14,15, 16,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
C23	2B1 "	37	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,39,40	"	"	"	"
C24	2A1 "	36	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 37,39,40	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C25	2BO Input breakdown current	31	3,4,5,6,8,9,11,12,14,15,16,17,18,23,24,25,26,29,30,32,34,36,37,39,40	0	1	mA	7
C26	2AO Input breakdown current	14	3,4,5,6,8,9,11,12,15,16,17,18,23,24,25,26,29,30,31,32,34,36,37,39,40	0	1	mA	7

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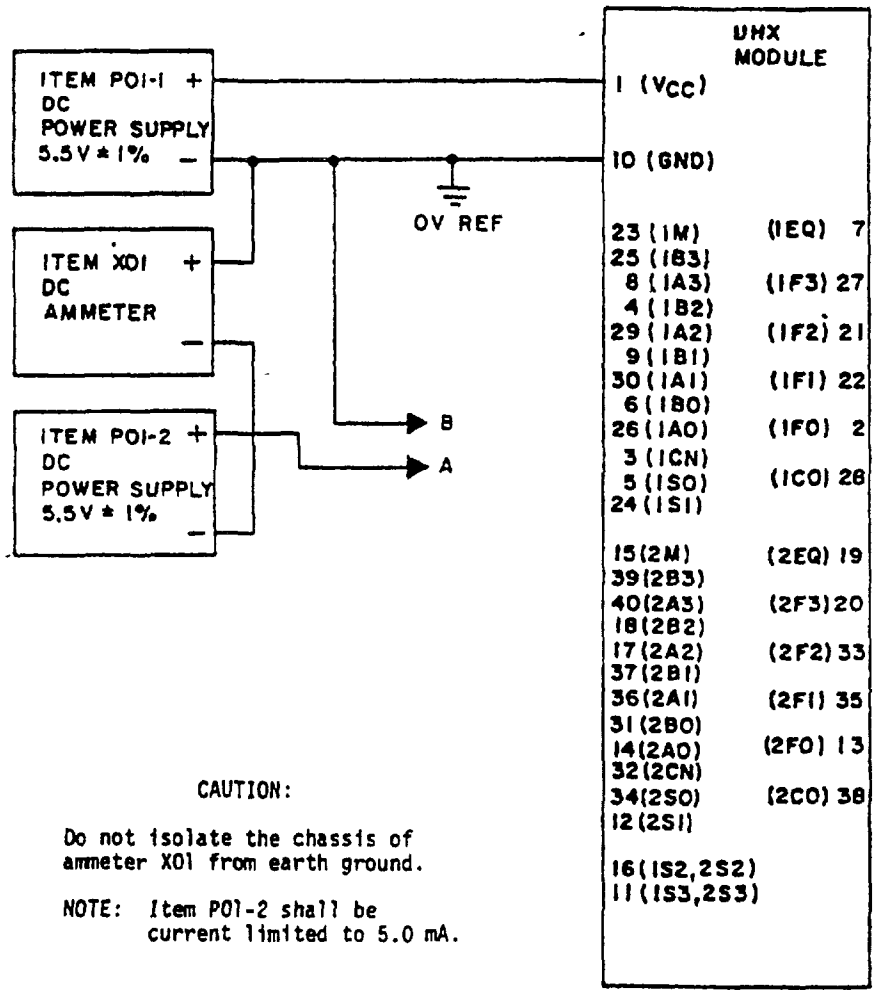


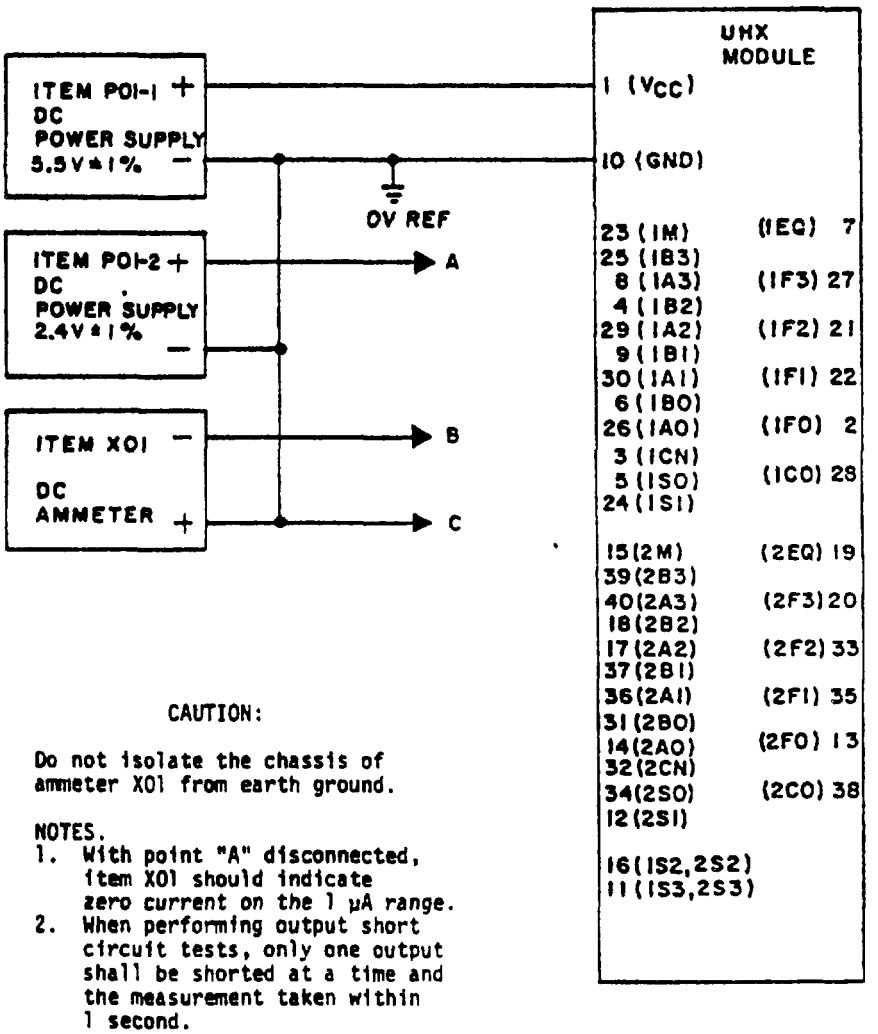
FIGURE 7. Input breakdown current test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1F3 Output short circuit current	15,16,23,11	27	4,6,9,12, 14,17,18, 24,25,31, 37,39,3,5, 8,26,29, 30,32,34, 36,40	-40	-100	mA	8
D2	1F2 "	"	21	"	"	"	"	"
D3	1F1 "	"	22	"	"	"	"	"
D4	1F0 "	"	2	"	"	"	"	"
D5	2F3 "	"	20	"	"	"	"	"
D6	2F2 "	"	33	"	"	"	"	"
D7	2F1 "	"	35	"	"	"	"	"
D8	2F0 "	"	13	"	"	"	"	"
D9	2EQ "	"	19	"	-3.5	-5.5	"	"
D10	1EQ "	"	7	"	-0.0	-0.5	"	"
D11	1C0 "	"	28	"	-40	-100	"	"
D12	2C0 "	"	38	"	-40	-100	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES.
1. With point "A" disconnected, item X01 should indicate zero current on the 1 µA range.
2. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 1 second.

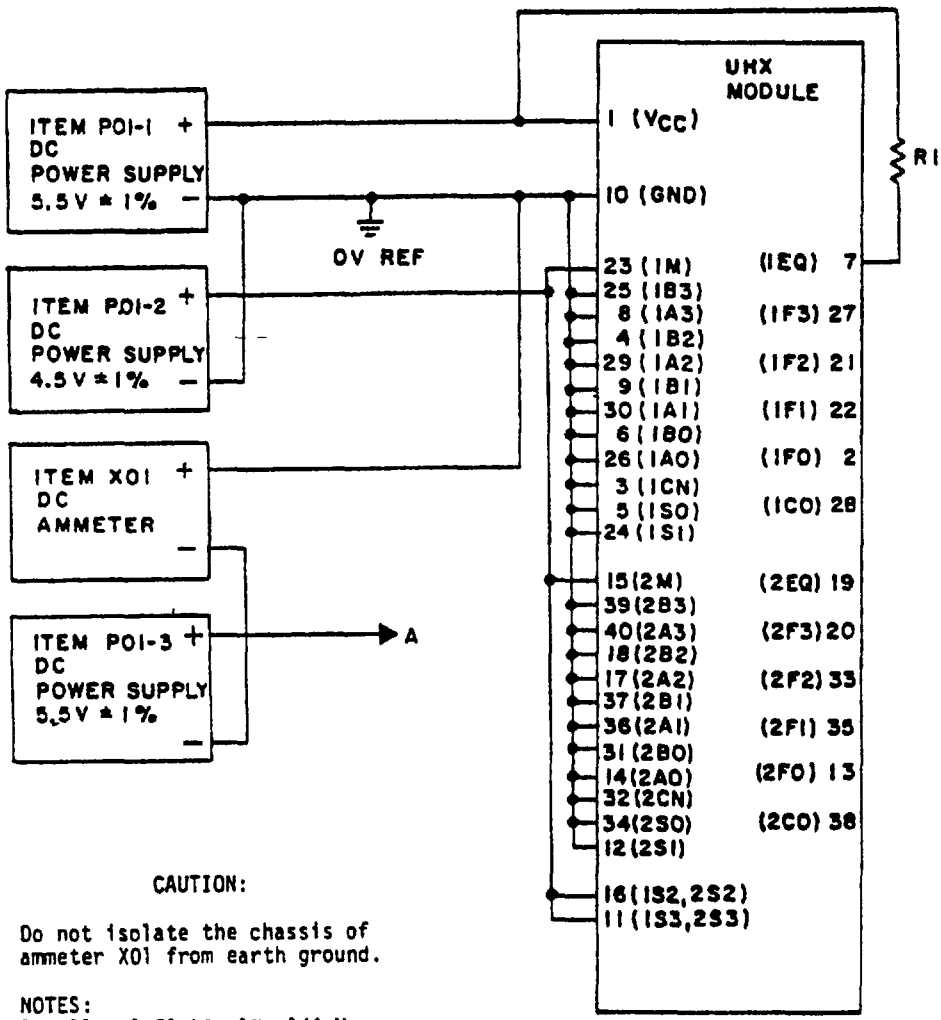
FIGURE 8. Output short circuit current test circuit for table VII.

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TABLE VIII. Output leakage current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1EQ Output leakage current	7	0	1	mA	9
E2	1F3 "	27	"	"	"	"
E3	1F2 "	21	"	"	"	"
E4	1F1 "	22	"	"	"	"
E5	1F0 "	2	"	"	"	"
E6	1C0 "	28	"	"	"	"
E7	2EQ "	19	"	"	"	"
E8	2F3 "	20	"	"	"	"
E9	2F2 "	33	"	"	"	"
E10	2F1 "	35	"	"	"	"
E11	2F0 "	13	"	"	"	"
E12	2C0 "	38	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 1.21 kΩ ±1%, 1/4 W.
2. Item PO1-3 shall be current limited to 5.0 mA.

FIGURE 9. Output leakage current test circuit for table VIII.

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TABLE IX. Input clamp diode voltage test.

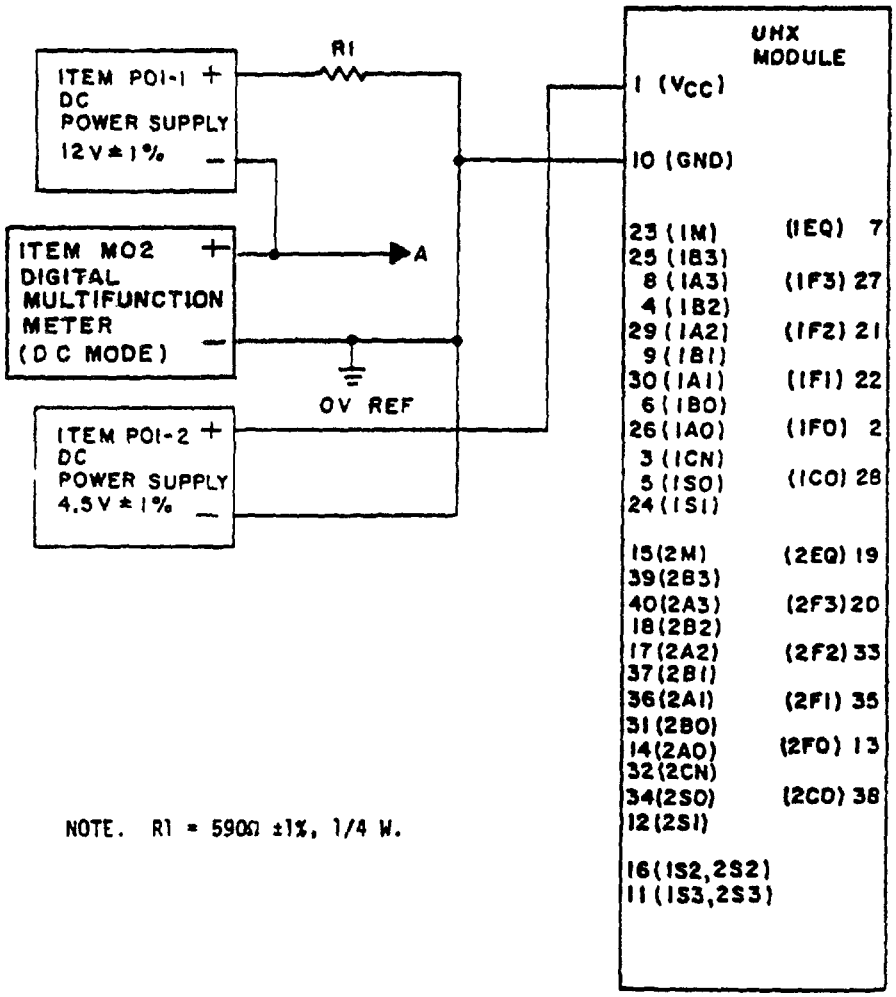
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	1B3	Input clamp diode voltage	25	-0.4	-1.2	V	10
F2	1A3	"	8	"	"	"	"
F3	1B2	"	4	"	"	"	"
F4	1A2	"	29	"	"	"	"
F5	1B1	"	9	"	"	"	"
F6	1A1	"	30	"	"	"	"
F7	1B0	"	6	"	"	"	"
F8	1A0	"	26	"	"	"	"
F9	1CN	"	3	"	"	"	"
F10	1M	"	23	"	"	"	"
F11	1S0	"	5	"	"	"	"
F12	1S1	"	24	"	"	"	"
F13	1S2, 2S2	"	16	"	"	"	"
F14	1S3, 2S3	"	11	"	"	"	"
F15	2S0	"	34	"	"	"	"
F16	2S1	"	12	"	"	"	"
F17	2M	"	15	"	"	"	"
F18	2CN	"	32	"	"	"	"
F19	2B3	"	39	"	"	"	"
F20	2A3	"	40	"	"	"	"
F21	2B2	"	18	"	"	"	"

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TABLE IX. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F22	2A2 Input clamp diode voltage	17	-0.4	-1.2	V	10
F23	2B1 "	37	"	"	"	"
F24	2A1 "	36	"	"	"	"
F25	2B0 "	31	"	"	"	"
F26	2A0 "	14	"	"	"	"

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NOTE. R1 = 590Ω ±1%, 1/4 W.

* FIGURE 10. Input clamp diode voltage test circuit for table IX.

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TABLE X. Output voltage (high-level) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G1	1F3 Output voltage, high level	8	3,4,5,6,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	27	27	Open	2.5	4.5	V	11
G2	1F2 "	3,29	4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,30, 31,32,34, 36,37,39,40	21	21	"	"	"	"	"
G3	1F1 "	3,30	4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 31,32,34, 36,37,39,40	22	22	"	"	"	"	"
G4	1F0 "	3,26	4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,29,30, 31,32,34, 36,37,39,40	2	2	"	"	"	"	"
G5	1C0 "	24,16, 3,25, 4,9,6	5,8,11,12, 14,15,17,18, 23,26,29,30, 31,32,34, 36,37,39,40	28	28	"	"	"	"	"
G6	2C0 "	12,16, 32,39, 18,37, 31	3,4,5,6,8,9, 11,14,15,17, 23,24,25,26, 29,30,34,36, 40	38	38	"	"	"	"	"

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TABLE X. Output voltage (high-level) test - Continued.

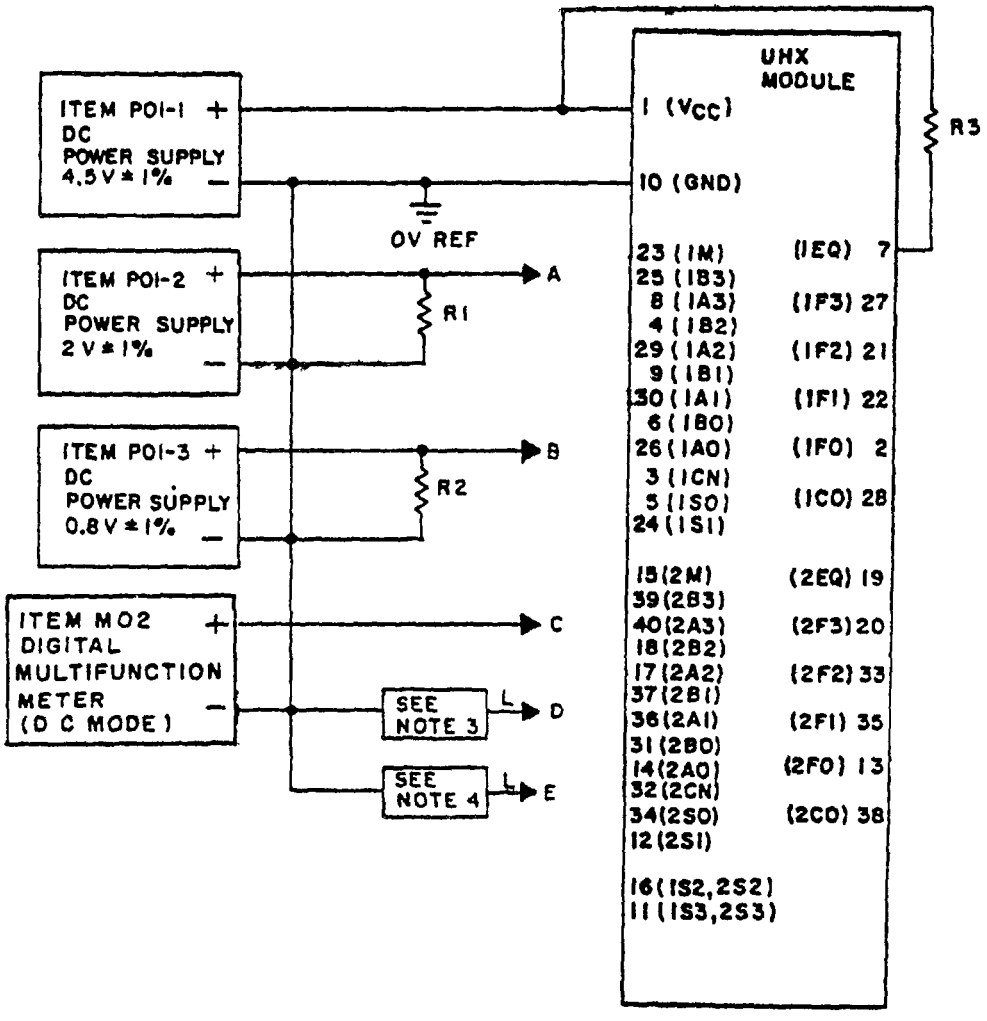
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G7	2F3 Output voltage, high level	32,40	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,34,36, 37,39	20	20	Open	2.5	4.5	V	11
G8	2F2 "	32,18	3,4,5,6,8,9, 11,12,14,15, 16,17,23, 24,25,26,29, 30,31,34,36, 37,39,40	33	33	"	"	"	"	"
G9	2F1 "	32,36	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,34, 37,39,40	35	35	"	"	"	"	"
G10	2F0 "	32,14	3,4,5,6,8,9, 11,12,15, 16,17,18,23, 24,25,26,29, 30,31,34,36, 37,39,40	13	13	"	"	"	"	"
G11	1EQ "	3,16 24	4,5,6,8,9, 11,12,14,15, 17,18,23,25, 26,29,30,31, 32,34,36, 37,39,40	7	Open	7	"	"	"	"
G12	2EQ "	32,12 26	3,4,5,6,8,9, 11,14,15,17, 18,23,24,25, 26,29,30,31, 34,36,37,39, 40	19	Open	19	"	"	"	"

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TABLE XI. Output voltage (low-level) test .

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
H1	1F3 Output voltage, low level	5,8,11, 14,17, 26,29, 30,34, 36,40	3,4,6,9,12, 15,16,18,23, 24,25,31,32, 37,39	27	27	Open	0.0	0.5	V	11
H2	1F2 "	"	"	21	21	"	"	"	"	"
H3	1F1 "	"	"	22	22	"	"	"	"	"
H4	1F0 "	"	"	2	2	"	"	"	"	"
H5	1C0 "	"	"	28	28	"	"	"	"	"
H6	2C0 "	"	"	38	38	"	"	"	"	"
H7	2F3 "	"	"	20	20	"	"	"	"	"
H8	2F2 "	"	"	33	33	"	"	"	"	"
H9	2F1 "	"	"	35	35	"	"	"	"	"
H10	2F0 "	"	"	13	13	"	"	"	"	"
H11	1EQ "	"	"	7	Open	7	"	"	"	"
H12	2EQ "	"	"	19	Open	19	"	"	"	"

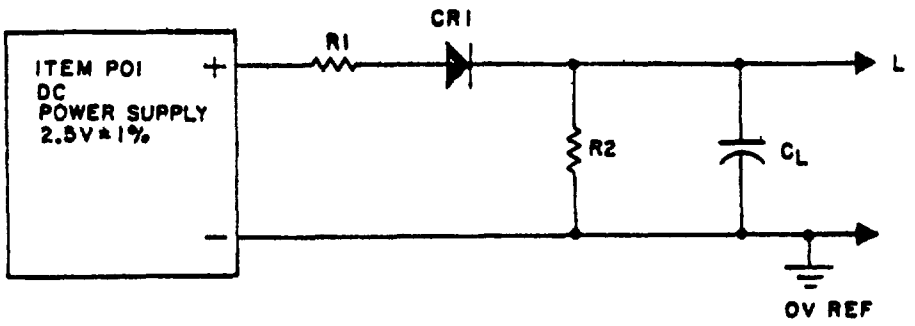
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 1.21 kΩ ±1%, 1/4 W.
3 Use load circuit 1, figure 12.
4. Use load circuit 2, figure 12.

* FIGURE 11. Output voltage (high-level and low-level) test circuit for tables X and XI.

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Load circuit 1:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. C_L is not required for dc electrical tests E1-E12 and F1-F12.
6. The maximum lead length from point L to the appropriate module pin shall be 1 inch (25.4 mm).

Load circuit 2:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% with point L held at 0.5 V.
3. R2 = 9.76 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. C_L is not required for dc electrical tests E1-E12 and F1-F12.
6. The maximum lead length from point L to the appropriate module pin shall be 1 inch (25.4 mm).

FIGURE 12. Logic load circuit.

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TABLE XII. Input current (low-level) test.

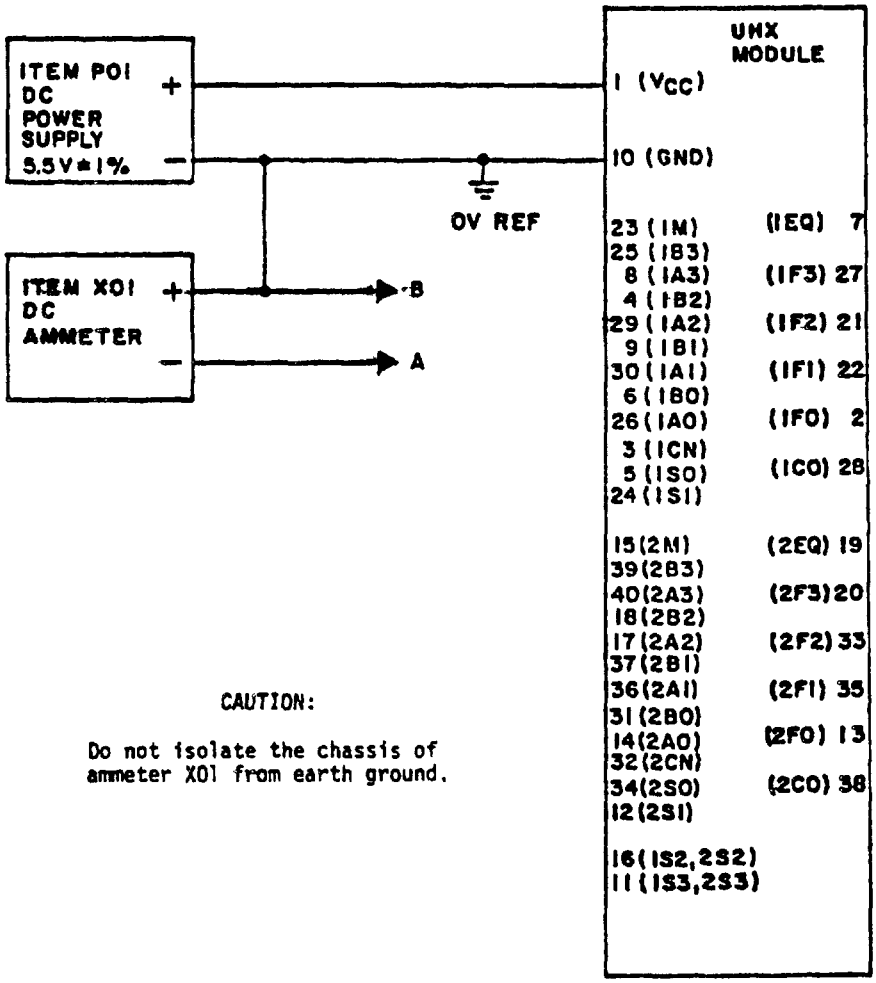
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1M Input current, low level	23	Open	-0.4	-2.2	mA	13
J2	2M "	15	"	-0.4	-2.2	"	"
J3	1B3 "	25	"	-1.2	-6.6	"	"
J4	1A3 "	8	"	"	"	"	"
J5	1B2 "	4	"	"	"	"	"
J6	1A2 "	29	"	"	"	"	"
J7	1B1 "	9	"	"	"	"	"
J8	1A1 "	30	"	"	"	"	"
J9	1B0 "	6	"	"	"	"	"
J10	1A0 "	26	"	"	"	"	"
J11	2B3 "	39	"	"	"	"	"
J12	2A3 "	40	"	"	"	"	"
J13	2B2 "	18	"	"	"	"	"
J14	2A2 "	17	"	"	"	"	"
J15	2B1 "	37	"	"	"	"	"
J16	2A1 "	36	"	"	"	"	"
J17	2B0 "	31	"	"	"	"	"
J18	2A0 "	14	"	"	"	"	"
J19	1S0 "	5	"	-1.6	-8.8	"	"
J20	1S1 "	24	"	-1.6	-8.8	"	"
J21	2S0 "	34	"	-1.6	-8.8	"	"

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J22	2S1	Input current, low level	12	Open	-1.6	-8.8	mA	13
J23	1S2, 2S2	"	16	"	-3.2	-17.6	"	"
J24	1S3, 2S3	"	11	"	-3.2	-17.6	"	"
J25	1CN	"	3	5,8,11,16,23, 24,26,29,30	-2	-11	"	"
J26	2CN	"	32	11,12,14,15, 16,17,34,36, 40	-2	-11	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 13. Input current (low-level) test circuit for table XII

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1M Input current, high level	23	3,4,5,6,8,9, 11,12,14,15, 16,17,18,24, 25,26,29,30, 31,32,34,36, 37,39,40	0	50	µA	14
K2	2M "	15	3,4,5,6,8,9, 11,12,14,16, 17,18,23,24, 25,26,29,30, 31,32,34,36, 37,39,40	"	50	"	"
K3	1B3 "	25	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,26,29,30, 31,32,34,36, 37,39,40	"	150	"	"
K4	1A3 "	8	3,4,5,6,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
K5	1B2 "	4 "	3,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
K6	1A2 "	29	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,30, 31,32,34,36, 37,39,40	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K7	IB1 Input current, high level	9	3,4,5,6,8, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	0	150	µA	14
K8	1A1 "	30	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 31,32,34, 36,37,39,40	"	"	"	"
K9	1B0 "	6	3,4,5,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
K10	1A10 "	26	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,29, 30,31,32,34, 36,37,39,40	"	"	"	"
K11	2B3 "	39	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,40	"	"	"	"
K12	2A3 "	40	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K13	2B2 Input current, high level	18	3,4,5,6,8,9, 11,12,14,15, 16,17,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	0	150	µA	14
K14	2A2 "	17	3,4,5,6,8,9, 11,12,14,15, 16,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
K15	2B1 "	37	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,39,40	"	"	"	"
K16	2A1 "	36	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 37,39,40	"	"	"	"
K17	2B0 "	31	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,32,34,36, 37,39,40	"	"	"	"
K18	2A0 "	14	3,4,5,6,8,9, 11,12,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K19	1S0	Input current, high level	5	3,4,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	0	200	µA	14
K20	1S1	"	24	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
K21	2S0	"	34	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32, 36,37,39,40	"	"	"	"
K22	2S1	"	12	3,4,5,6,8,9, 11,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	"	"	"
K23	1S2, 2S2	"	16	3,4,5,6,8,9, 11,12,14,15, 17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	400	"	"
K24	1S3, 2S3	"	11	3,4,5,6,8,9, 12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	"	400	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K25	1CN Input current, high level	3	4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,32,34, 36,37,39,40	0	250	µA	14
K26	2CN Input current, high level	32	3,4,5,6,8,9, 11,12,14,15, 16,17,18,23, 24,25,26,29, 30,31,34, 36,37,39,40	0	250	µA	14

TABLE XIV. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Power supply current, high level	C	Open	A	1	440	mA	15
L2	Power supply current, low level	B	A	Open	1	440	mA	15

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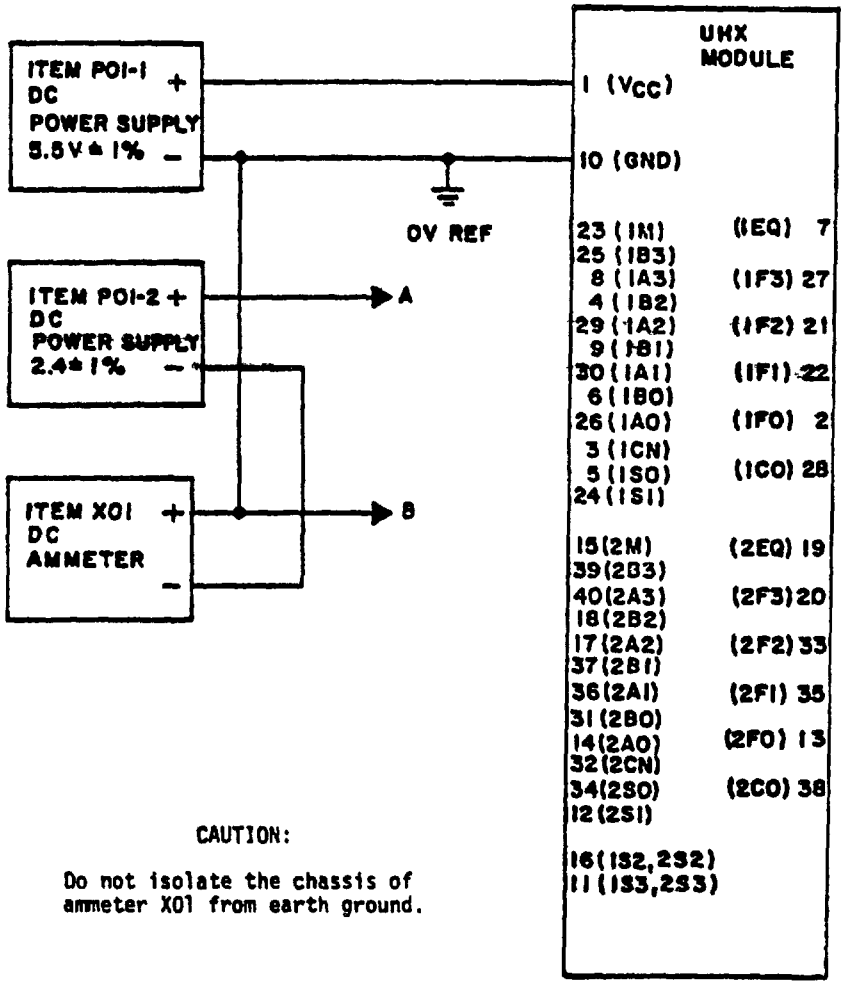


FIGURE 14. Input current (high-level) test circuit for table XIII.

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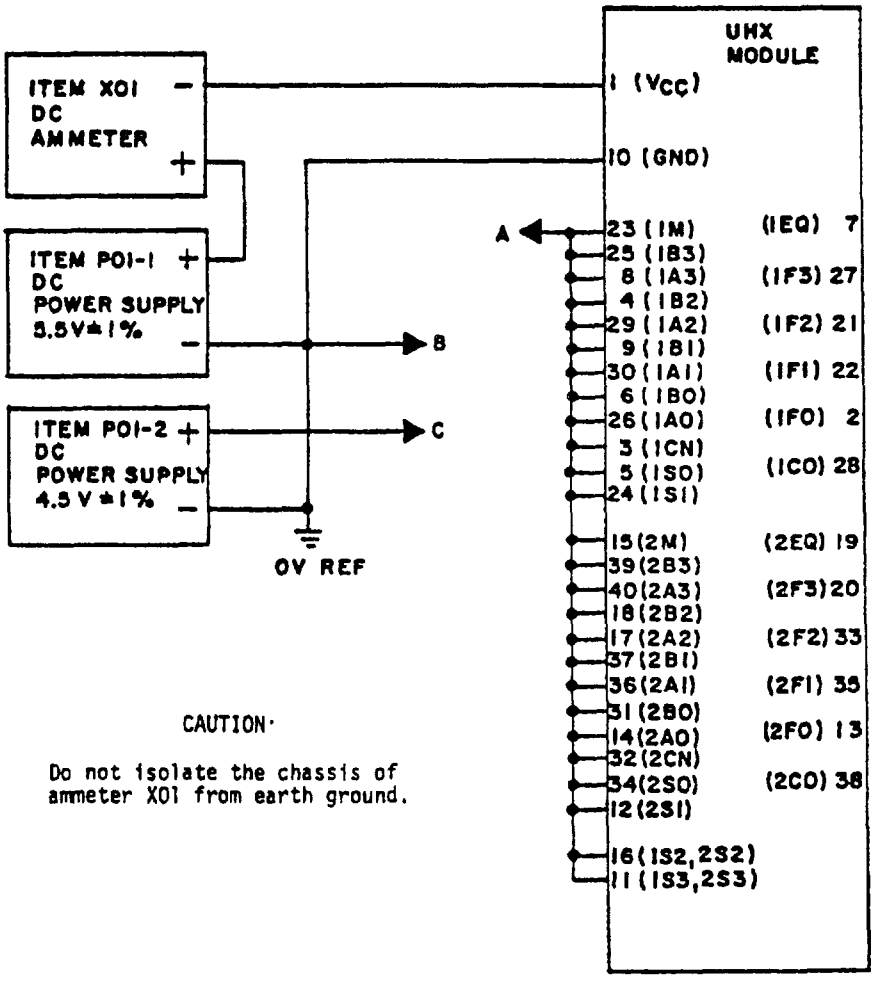


FIGURE 15. Power supply current (high-level and low-level) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1F3 Transition time (t_{TLH}), C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36, 40	3,32	27	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	2	15	ns	16,17
M2	1F2 "	"	"	21	"	"	"	"	"
M3	1F1 "	"	"	22	"	"	"	"	"
M4	1F0 "	"	"	2	"	"	"	"	"
M5	1C0 "	"	"	28	"	"	"	"	"
M6	2C0 "	"	"	38	"	"	"	"	"
M7	2F3 "	"	"	20	"	"	"	"	"
M8	2F2 "	"	"	33	"	"	"	"	"
M9	2F1 "	"	"	35	"	"	"	"	"
M10	2F0 "	"	"	13	"	"	"	"	"
M11	1F3 Transition time (t_{TLH}), C_L = 150 pF	"	"	27	"	"	30	"	"
M12	1F2 "	"	"	21	"	"	"	"	"
M13	1F1 "	"	"	22	"	"	"	"	"
M14	1F0 "	"	"	2	"	"	"	"	"
M15	1C0 "	"	"	28	"	"	"	"	"
M16	2C0 "	"	"	38	"	"	"	"	"
M17	2F3 "	"	"	20	"	"	"	"	"
M18	2F2 "	"	"	33	"	"	"	"	"
M19	2F1 "	"	"	35	"	"	"	"	"
M20	2F0 "	"	"	13	"	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M21	1EQ Transition time (t_{TLH}), C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36, 40	3,32	7	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	10	40	ns	16,17
M22	2EQ Transition time (t_{TLH}), C_L = 50 pF	"	"	19	"	"	40	"	"
M23	1EQ Transition time (t_{TLH}), C_L = 150 pF	"	"	7	"	"	65	"	"
M24	2EQ Transition time (t_{TLH}), C_L = 150 pF	"	"	19	"	"	65	"	"
N1	1F3 Transition time (t_{THL}), C_L = 50 pF	"	3,22	27	"	1	10	"	"
N2	1F2 "	"	"	21	"	"	"	"	"
N3	1F1 "	"	"	22	"	"	"	"	"
N4	1F0 "	"	"	2	"	"	"	"	"
N5	1C0 "	"	"	28	"	"	"	"	"
N6	2C0 "	"	"	38	"	"	"	"	"
N7	2F3 "	"	"	20	"	"	"	"	"
N8	2F2 "	"	"	33	"	"	"	"	"
N9	2F1 "	"	"	35	"	"	"	"	"
N10	2F0 "	"	"	13	"	"	"	"	"
N11	1F3 Transition time (t_{THL}), C_L = 150 pF	"	"	27	"	2	20	"	"
N12	1F2 "	"	"	21	"	"	"	"	"
N13	1F1 "	"	"	22	"	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N14	1FO Transition time (t_{THL}), C_L = 150 pF	5,8,11,14, 17,26,29, 30,34,36, 40	3,22	2	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	2	20	ns	16,17
N15	1CO "	"	3,22	28	"	"	"	"	"
N16	2CO "	"	3,32	38	"	"	"	"	"
N17	2F3 "	"	"	20	"	"	"	"	"
N18	2F2 "	"	"	33	"	"	"	"	"
N19	2F1 "	"	"	35	"	"	"	"	"
N20	2FO "	"	"	13	"	"	"	"	"
N21	1EQ Transition time (t_{THL}), C_L = 50 pF	"	"	7	"	1	10	"	"
N22	2EQ Transition time (t_{THL}), C_L = 50 pF	"	"	19	"	1	10	"	"
N23	1EQ Transition time (t_{THL}), C_L = 150 pF	"	"	7	"	2	20	"	"
N24	2EQ Transition time (t_{THL}), C_L = 150 pF	"	"	19	"	2	20	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1A0 to 1F0 Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	5,34,30, 29,11,3,8, 6,32	26	2	24,12,9, 4,16,23, 25,15	5	21	ns	16,17
P2	1A1 to 1F1 "	5,34,26, 29,11,3,8, 9,32	30	22	24,12, 6,4,16, 23,25,15	"	"	"	"
P3	1A2 to 1F2 "	5,34,26, 30,11,3,8, 4,32	29	21	24,12,6, 9,16, 23,25,15	"	"	"	"
P4	1A3 to 1F3 "	5,34,26, 30,11,3, 29,25,32	8	27	24,12,6, 9,16,23, 4,15	"	"	"	"
P5	2A0 to 2F0 "	5,34,36, 17,11,3, 40,31,32	14	13	24,12,37, 18,16,23, 39,15	"	"	"	"
P6	2A1 to 2F1 "	5,34,14, 17,11,3, 40,37,32	36	35	24,12,31, 18,16,23, 39,15	"	"	"	"
P7	2A2 to 2F2 "	5,34,14, 36,11,3, 40,18,32	17	33	24,12,31, 37,16,23, 39,15	"	"	"	"
P8	2A3 to 2F3 "	5,34,14, 36,11,3, 17,39,32	40	20	24,12,31, 37,16,23, 18,15	"	"	"	"
P9	1A0 to 1F0 Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	5,34,30, 29,11,3, 8,6,32	26	2	24,12,9, 4,16,23, 25,15	8	24	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P10	1A1 to 1F1 Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	5,34,26, 29,11,3, 8,9,32	30	22	24,12,6, 4,16,23, 25,15	8	24	ns	16,17
P11	1A2 to 1F2 "	5,34,26, 30,11,3, 8,4,32	29	21	24,12,6, 9,16,23, 25,15	"	"	"	"
P12	1A3 to 1F3 "	5,34,26, 30,11,3, 29,25,32	8	27	24,12,6, 9,16,23, 4,15	"	"	"	"
P13	2A0 to 2F0 "	5,34,36, 17,11,3, 40,31,32	14	13	24,12,37, 18,16,23, 39,15	"	"	"	"
P14	2A1 to 2F1 "	5,34,14, 17,11,3, 40,37,32	36	35	24,12,31, 18,16,23, 39,15	"	"	"	"
P15	2A2 to 2F2 "	5,34,14, 36,11,3, 40,18,32	17	33	24,12,31, 37,16,23, 39,15	"	"	"	"
P16	2A3 to 2F3 "	5,34,14, 36,11,3, 17,39,32	40	20	24,12,31, 37,16,23, 18,15	"	"	"	"
P17	1B0 to 1F0 Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	5,34,11, 3,32	6	2	24,12,8, 29,16,23, 30,26,15	5	21	"	"
P18	1B1 to 1F1 "	"	9	22	"	"	"	"	"
P19	1B2 to 1F2 "	"	4	21	"	"	"	"	"
P20	1B3 to 1F3 "	"	25	27	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P21	2B0 to 2F0 Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	5,34,11, 3,32	31	13	24,12,40, 17,16,23, 36,14,15	5	21	ns	16,17
P22	2B1 to 2F1 "	"	37	35	"	"	"	"	"
P23	2B2 to 2F2 "	"	18	33	"	"	"	"	"
P24	2B3 to 2F3 "	"	39	20	"	"	"	"	"
P25	1B0 to 1F0 Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	"	6	2	24,12,8, 29,16,23, 30,26,15	8	24	"	"
P26	1B1 to 1F1 "	"	9	22	"	"	"	"	"
P27	1B2 to 1F2 "	"	4	21	"	"	"	"	"
P28	1B3 to 1F3 "	"	25	27	"	"	"	"	"
P29	2B0 to 2F0 "	"	31	13	24,12,40, 17,16,23, 36,14,15	"	"	"	"
P30	2B1 to 2F1 "	"	37	35	"	"	"	"	"
P31	2B2 to 2F2 "	"	18	33	"	"	"	"	"
P32	2B3 to 2F3 "	"	39	20	"	"	"	"	"
P33	1CN to 1F0 Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36	3	2	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	3	17	"	"
P34	1CN to 1F1 Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36	3	22	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	3	17	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P35	1CN to 1F2 Propagation delay time (t_{PLH}), sum mode, C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36	3	21	4,6,9,12, 15,16,18, 23,24,25 31,37,39	3	17	ns	16,17
P36	1CN to 1F3 "	"	3	27	"	"	"	"	"
P37	2CN to 2F0 "	"	32	13	"	"	"	"	"
P38	2CN to 2F1 "	"	"	35	"	"	"	"	"
P39	2CN to 2F2 "	"	"	33	"	"	"	"	"
P40	2CN to 2F3 "	"	"	20	"	"	"	"	"
P41	1CN to 1F0 Propagation delay time (t_{PLH}), sum mode, C_L = 150 pF	"	3	2	"	5	20	"	"
P42	1CN to 1F1 "	"	"	22	"	"	"	"	"
P43	1CN to 1F2 "	"	"	21	"	"	"	"	"
P44	1CN to 1F3 "	"	"	27	"	"	"	"	"
P45	2CN to 2F0 "	"	32	13	"	"	"	"	"
P46	2CN to 2F1 "	"	"	35	"	"	"	"	"
P47	2CN to 2F2 "	"	"	33	"	"	"	"	"
P48	2CN to 2F3 "	"	"	20	"	"	"	"	"
P49	1CN to 1C0 Propagation delay time (t_{PLH}), sum mode C_L = 50 pF	"	3	28	"	3	12	"	"
P50	2CN to 2C0 Propagation delay time (t_{PLH}), sum mode C_L = 50 pF	"	32	38	"	3	12	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P51	1CN to 1CO Propagation delay time (t_{PLH}), sum mode C_L = 150 pF	5,8,11,14, 17,26,29, 30,34,36	3	28	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	5	15	ns	16,17
P52	2CN to 2CO Propagation delay time (t_{PLH}), sum mode C_L = 150 pF	5,8,11,14, 17,26,29, 30,34,36	32	38	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	"	15	"	"
P53	1AO to 1CO Propagation delay time (t_{PLH}), sum mode C_L = 50 pF	5,34,11, 9,4,25	26	28	24,12,16, 23,15,6, 30,29,8,3	"	20	"	"
P54	1A1 to 1CO "	5,34,11, 6,4,25	30	"	24,12,16, 23,15,9, 26,29,8,3	"	"	"	"
P55	1A2 to 1CO "	5,34,11, 6,9,25	29	"	24,12,16, 23,15,4, 26,30,8,3	"	"	"	"
P56	1A3 to 1CO "	5,34,11, 6,9,4	8	"	24,12,16, 23,15,25, 26,30,29, 3	"	"	"	"
P57	2AO to 2CO "	5,34,11, 37,18,39	14	38	24,12,16, 23,15,31, 36,17,40, 32	"	"	"	"
P58	2A1 to 2CO "	5,34,11, 31,18,39	36	38	24,12,16, 23,15,37, 14,17,40, 32	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P59	2A2 to 2CO Propagation delay time (t_{PLH}), sum mode C_L = 50 pF	5,34,11, 31,37,39	17	38	24,12,16, 23,15,18, 14,36,40, 32	5	20	ns	16,17
P60	2A3 to 2CO Propagation delay time (t_{PLH}), sum mode C_L = 50 pF	5,34,11, 31,37,18	40	38	24,12,16, 23,15,39, 14,36,17, 32	5	20	"	"
P61	1AO to 1CO Propagation delay time (t_{PLH}), sum mode C_L = 150 pF	5,34,11, 9,4,25	26	28	24,12,16, 23,15,6, 30,29,8, 3	8	23	"	"
P62	1A1 to 1CO "	5,34,11, 6,4,25	30	"	24,12,16, 23,15,9, 26,29,8, 3	"	"	"	"
P63	1A2 to 1CO "	5,34,11, 6,9,25	29	"	24,12,16, 23,15,4, 26,30,8, 3	"	"	"	"
P64	1A3 to 1CO "	5,34,11, 6,9,4	8	"	24,12,16, 23,15,25, 26,30,29, 3	"	"	"	"
P65	2AO to 2CO "	5,34,11, 37,18,39	14	38	24,12,16, 23,15,31, 36,17,40, 32	"	"	"	"
P66	2A1 to 2CO "	5,34,11, 31,18,39	36	38	24,12,16, 23,15,37, 14,17,40, 32	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P67	2A2 to 2C0 Propagation delay time (t_{PLH}), sum mode C_L = 150 pF	5,34,11, 31,37,39	17	38	24,12,16, 23,15,18, 14,36,40, 32	8	23	ns	16,17
P68	2A3 to 2C0 Propagation delay time (t_{PLH}), sum mode C_L = 150 pF	5,34,11, 31,37,18	40	38	24,12,16, 23,15,39, 14,36,17, 32	8	23	"	"
P69	1B0 to 1C0 Propagation delay time (t_{PLH}), sum mode C_L = 50 pF	5,34,11, 9,4,25	6	28	24,12,16, 23,15,26, 30,29,8, 3	5	20	"	"
P70	1B1 to 1C0 "	5,34,11, 6,4,25	9	"	"	"	"	"	"
P71	1B2 to 1C0 "	5,34,11, 6,9,25	4	"	"	"	"	"	"
P72	1B3 to 1C0 "	5,34,11, 6,9,4	25	"	"	"	"	"	"
P73	2B0 to 2C0 "	5,34,11, 37,18,39	31	38	24,12,16 23,15,14, 36,17,40, 32	"	"	"	"
P74	2B1 to 2C0 "	5,34,11, 31,18,39	37	"	"	"	"	"	"
P75	2B2 to 2C0 "	5,34,11, 31,37,39	18	"	"	"	"	"	"
P76	2B3 to 2C0 "	5,34,11, 31,37,18	39	"	"	"	"	"	"
P77	1B0 to 1C0 Propagation delay time (t_{PLH}), sum mode C_L = 150 pF	5,34,11, 9,4,25	6	28	24,12,16, 23,15,26, 30,29,8,3	8	23	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P78	1B1 to 1C0 Propagation delay time (t_{PLH}), sum mode C_L = 150 pF	5,34,11, 6,4,25	9	28	24,12,16, 23,15,26, 30,29,8,3	8	23	ns	16,17
P79	1B2 to 1C0 "	5,34,11, 6,9,25	4	28	"	"	"	"	"
P80	1B3 to 1C0 "	5,34,11, 6,9,4	25	28	"	"	"	"	"
P81	2B0 to 2C0 "	5,34,11, 37,18,39	31	38	24,12,16, 23,15,14, 36,17,40, 32	"	"	"	"
P82	2B1 to 2C0 "	5,34,11, 31,18,39	37	"	"	"	"	"	"
P83	2B2 to 2C0 "	5,34,11, 31,37,39	18	"	"	"	"	"	"
P84	2B3 to 2C0 "	5,34,11, 31,37,18	39	"	"	"	"	"	"
P85	1A0 to 1F0 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12,16, 30,29,8	26	2	5,34,11, 23,15,6, 9,4,25,3	5	24	"	"
P86	1A1 to 1F1 "	24,12,16, 26,29,8	30	22	"	"	"	"	"
P87	1A2 to 1F2 "	24,12,16, 26,30,8	29	21	"	"	"	"	"
P88	1A3 to 1F3 "	24,12,16, 26,30,29	8	27	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P89	2A0 to 2F0 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12,16, 36,17,40	14	13	5,34,11, 23,15,31, 37,18,39, 32	5	24	ns	16,17
P90	2A1 to 2F1 "	24,12,16, 14,17,40	36	35	"	"	"	"	"
P91	2A2 to 2F2 "	24,12,16, 14,36,40	17	33	"	"	"	"	"
P92	2A3 to 2F3 "	24,12,16, 14,36,17	40	20	"	"	"	"	"
P93	1A0 to 1F0 Propagation delay time (t_{PLH}), difference mode C_L = 150 pF	24,12,16, 30,29,8	26	2	5,34,11, 23,15,6, 9,4,25,3	8	27	"	"
P94	1A1 to 1F1 "	24,12,16, 26,29,8	30	22	"	"	"	"	"
P95	1A2 to 1F2 "	24,12,16, 26,30,8	29	21	"	"	"	"	"
P96	1A3 to 1F3 "	24,12,16, 26,30,29	8	27	"	"	"	"	"
P97	2A0 to 2F0 "	24,12,16, 36,17,40	14	13	5,34,11, 23,15,31, 37,18,39, 32	"	"	"	"
P98	2A1 to 2F1 "	24,12,16, 14,17,40	36	35	5,34,11, 23,15,31, 37,18,39, 32	"	"	"	"
P99	2A2 to 2F2 "	24,12,16, 14,36,40	17	33	5,34,11, 23,15,31, 37,18,39, 32	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P100	2A3 to 2F3 Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	24,12,16, 14,36,17	40	20	5,34,11, 23,15,31, 37,18,39, 32	8	27	ns	16,17
P101	1B0 to 1F0 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12,16, 26,30,29 8	6	2	5,34,11, 23,15,9, 4,25,3	5	24	"	"
P102	1B1 to 1F1 "	"	9	22	5,34,11, 23,15,6, 4,25,3	"	"	"	"
P103	1B2 to 1F2 "	"	4	21	5,34,11, 23,15,6, 9,25,3	"	"	"	"
P104	1B3 to 1F3 "	"	25	27	5,34,11, 23,15,6, 9,4,3	"	"	"	"
P105	2B0 to 2F0 "	24,12,16, 14,36,17, 40	31	13	5,34,11, 23,15,37, 18,39,32	"	"	"	"
P106	2B1 to 2F1 "	"	37	35	5,34,11, 23,15,31, 18,39,32	"	"	"	"
P107	2B2 to 2F2 "	"	18	33	5,34,11, 23,15,31, 37,39,32	"	"	"	"
P108	2B3 to 2F3 "	"	39	20	5,34,11, 23,15,31, 37,18,32	"	"	"	"
P109	1B0 to 1F0 Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	24,12,16, 26,30,29, 8	6	2	5,34,11, 23,15,9, 4,25,3	8	27	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P110	1B1 to 1F1 Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	24,12,16, 26,30,29, 8	9	22	5,34,11, 23,15,6, 4,25,3	8	27	ns	16,17
P111	1B2 to 1F2 "	"	4	21	5,34,11, 23,15,6, 9,25,3	"	"	"	"
P112	1B3 to 1F3 "	"	25	27	5,34,11, 23,15,6, 9,4,3	"	"	"	"
P113	2B0 to 2F0 "	24,12,16, 14,36,17, 40	31	13	5,34,11, 23,15,37, 18,39,32	"	"	"	"
P114	2B1 to 2F1 "	"	37	35	5,34,11, 23,15,31, 18,39,32	"	"	"	"
P115	2B2 to 2F2 "	"	18	33	5,34,11, 23,15,31, 37,39,32	"	"	"	"
P116	2B3 to 2F3 "	"	39	20	5,34,11, 23,15,31, 37,18,32	"	"	"	"
P117	1CN to 1F0 Propagation delay time (t_{PLH}), difference mode C_L = 50 pF	24,12,16, 26,30,29, 8,6,9,4, 25	3	2	5,34,11, 23,15	5	17	"	"
P118	1CN to 1F1 "	"	"	22	"	"	"	"	"
P119	1CN to 1F2 "	"	"	21	"	"	"	"	"
P120	1CN to 1F3 "	"	"	27	"	"	"	"	"
P121	2CN to 2F0 "	24,12,16, 14,36,17, 40,31,37, 18,39	32	13	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P122	2CN to 2F1 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12,16, 14,36,17, 40,31,37, 18,39	32	35	5,34,11, 23,15	5	17	ns	16,17
P123	2CN to 2F2 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12,16, 14,36,17, 40,31,37, 18,39	32	33	"	5	17	"	"
P124	2CN to 2F3 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12,16, 14,36,17, 40,31,37, 18,39	32	20	"	5	17	"	"
P125	1CN to 1F0 Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	24,12,16, 26,30,29, 8,6,9,4, 25	3	2	"	8	20	"	"
P126	1CN to 1F1 "	"	"	22	"	"	"	"	"
P127	1CN to 1F2 "	"	"	21	"	"	"	"	"
P128	1CN to 1F3 "	"	"	27	"	"	"	"	"
P129	2CN to 2F0 "	24,12,16, 14,36,17, 40,31,37, 18,39	32	13	"	"	"	"	"
P130	2CN to 2F1 "	"	"	35	"	"	"	"	"
P131	2CN to 2F2 "	"	"	33	"	"	"	"	"
P132	2CN to 2F3 "	"	"	20	"	"	"	"	"
P133	1A0 to 1C0 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12, 16,6	26	28	5,34,11, 23,15,30, 29,8,9,4, 25,3	5	27	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P134	1A1 to 1CO Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12, 16,9	30	28	5,34,11, 23,15,26, 29,8,6,4, 25,3	5	27	ns	16,17
P135	1A2 to 1CO "	24,12, 16,4	29	28	5,34,11, 23,15,26, 30,8,6,9, 25,3	"	"	"	"
P136	1A3 to 1CO "	24,12, 16,25	8	28	5,34,11, 23,15,26, 30,29,6, 9,4,3	"	"	"	"
P137	2A0 to 2CO "	24,12, 16,31	14	38	5,34,11, 23,15,36, 17,40,37, 18,39,32	"	"	"	"
P138	2A1 to 2CO "	24,12, 16,37	36	"	5,34,11, 23,15,14, 17,40,31, 18,39,32	"	"	"	"
P139	2A2 to 2CO "	24,12, 16,18	17	"	5,34,11, 23,15,14, 36,40,31, 37,39,32	"	"	"	"
P140	2A3 to 2CO "	24,12, 16,39	40	"	5,34,11, 23,15,14, 36,17,31, 37,18,32	"	"	"	"
P141	1A0 to 1CO Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	24,12, 16,6	26	28	5,34,11, 23,15,30, 29,8,9,4, 25,3	8	31	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P142	1A1 to 1C0 Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	24,12, 16,9	30	28	5,34,11, 23,15,26, 29,8,6,4, 25,3	8	31	ns	16,17
P143	1A2 to 1C0 "	24,12, 16,4	29	28	5,34,11, 23,15,26, 30,8,6,9, 25,3	"	"	"	"
P144	1A3 to 1C0 "	24,12, 16,25	8	28	5,34,11, 23,15,26, 30,29,6, 9,4,3	"	"	"	"
P145	2A0 to 2C0 "	24,12, 16,31	14	38	5,34,11, 23,15,36, 17,40,37, 18,39,32	"	"	"	"
P146	2A1 to 2C0 "	24,12, 16,37	36	"	5,34,11, 23,15,14, 17,40,31, 18,39,32	"	"	"	"
P147	2A2 to 2C0 "	24,12, 16,18	17	"	5,34,11, 23,15,14, 36,40,31, 37,39,32	"	"	"	"
P148	2A3 to 2C0 "	24,12, 16,39	40	"	5,34,11, 23,15,14, 36,17,31, 37,18,32	"	"	"	"
P149	1B0 to 1C0 Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12, 16	6	28	5,34,11, 23,15,26, 30,29,8, 9,4,25,3	5	27	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P150	1B1 to 1CO Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	24,12, 16	9	28	5,34,11, 23,15,26, 30,29,8, 6,4,25,3	5	27	ns	16,17
P151	1B2 to 1CO "	"	4	28	5,34,11, 23,15,26, 30,29,8, 6,9,25,3	"	"	"	"
P152	1B3 to 1CO "	"	25	28	5,34,11, 23,15,26, 30,29,8, 6,9,4,3	"	"	"	"
P153	2B0 to 2CO "	"	31	38	5,34,11, 23,15,14, 36,17,40, 37,18,39, 32	"	"	"	"
P154	2B1 to 2CO "	"	37	"	5,34,11, 23,15,14, 36,17,40, 31,18,39, 32	"	"	"	"
P155	2B2 to 2CO "	"	18	"	5,34,11, 23,15,14, 36,17,40, 31,37,39, 32	"	"	"	"
P156	2B3 to 2CO "	"	39	"	5,34,11, 23,15,14, 36,17,40, 31,37,18, 32	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P157	1B0 to 1C0 Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	24,12, 16	6	28	5,34,11, 23,15,26, 30,29,8, 9,4,25,3	8	31	ns	16,17
P158	1B1 to 1C0 "	"	9	"	5,34,11, 23,15,26, 30,29,8, 6,4,25,3	"	"	"	"
P159	1B2 to 1C0 "	"	4	"	5,34,11, 23,15,26, 30,29,8, 6,9,25,3	"	"	"	"
P160	1B3 to 1C0 "	"	25	"	5,34,11, 23,15,26, 30,29,8, 6,9,4,3	"	"	"	"
P161	2B0 to 2C0 "	"	31	38	5,34,11, 23,15,14, 36,17,40, 37,18,39, 32	"	"	"	"
P162	2B1 to 2C0 "	"	37	"	5,34,11, 23,15,14, 36,17,40, 31,18,39, 32	"	"	"	"
P163	2B2 to 2C0 "	"	18	"	5,34,11, 23,15,14, 36,17,40, 31,37,39, 32	"	"	"	"
P164	2B3 to 2C0 "	"	39	"	5,34,11, 23,15,14, 36,17,40, 31,37,18, 32	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P165	1A0 to 1EQ Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	3,12,4,6, 9,16,24, 25,8,32, 29,30	26	7	5,11,15, 23,34	10	42	ns	16,17
P166	1A1 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,29	30	"	"	"	"	"	"
P167	1A2 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,30	29	"	"	"	"	"	"
P168	1A3 to 1EQ "	3,12,4,6, 9,16,24, 25,26,32, 29,30	8	"	"	"	"	"	"
P169	2A0 to 2EQ "	3,12,18, 31,16,24 37,39,32, 17,36,40	14	19	"	"	"	"	"
P170	2A1 to 2EQ "	3,12,18, 31,16,24 37,39,32, 14,17,40	36	"	"	"	"	"	"
P171	2A2 to 2EQ "	3,12,18, 31,16,24 37,39,32, 14,36,40	17	"	"	"	"	"	"
P172	2A3 to 2EQ "	3,12,18, 31,16,24 37,39,32, 14,17,36	40	"	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P173	1A0 to 1EQ Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	3,12,4,6, 9,16,24, 25,8,32, 29,30	26	7	5,11,15, 23,34	10	45	ns	16,17
P174	1A1 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,29	30	"	"	"	"	"	"
P175	1A2 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,30	29	"	"	"	"	"	"
P176	1A3 to 1EQ "	3,12,4,6, 9,16,24, 25,26,32, 29,30	8	"	"	"	"	"	"
P177	2A0 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 17,36,40	14	19	"	"	"	"	"
P178	2A1 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 14,17,40	36	"	"	"	"	"	"
P179	2A2 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 14,36,40	17	"	"	"	"	"	"
P180	2A3 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 14,17,36	40	"	"	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P181	1B0 to 1EQ Propagation delay time (t_{PLH}), difference mode, C_L = 50 pF	3,12,16, 24,32	6	7	5,11,8, 26,15,23, 29,30,34, 4,9,25	10	42	ns	16,17
P182	1B1 to 1EQ "	"	9	"	5,11,8, 26,15,23, 29,30,34, 4,6,25	"	"	"	"
P183	1B2 to 1EQ "	"	4	"	5,11,8, 26,15,23, 29,30,34, 6,9,25	"	"	"	"
P184	1B3 to 1EQ "	"	25	"	5,11,8, 26,15,23, 29,30,34, 4,6,9	"	"	"	"
P185	2B0 to 2EQ "	"	31	19	5,11,14, 17,15,23, 36,40,34, 18,37,39	"	"	"	"
P186	2B1 to 2EQ "	"	37	"	5,11,14, 17,15,23, 36,40,34, 18,31,39	"	"	"	"
P187	2B2 to 2EQ "	"	18	"	5,11,14, 17,15,23, 36,40,34, 31,37,39	"	"	"	"
P188	2B3 to 2EQ "	"	39	"	5,11,14, 17,15,23, 36,40,34, 18,31,37	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P189	1B0 to 1EQ Propagation delay time (t_{PLH}), difference mode, C_L = 150 pF	3,12,16, 24,32	6	7	5,11,8, 26,15,23, 29,30,34, 4,9,25	10	45	ns	16,17
P190	1B1 to 1EQ "	"	9	"	5,11,8, 26,15,23, 29,30,34, 4,6,25	"	50	"	"
P191	1B2 to 1EQ "	"	4	"	5,11,8, 26,15,23, 29,30,34, 6,9,25	"	50	"	"
P192	1B3 to 1EQ "	"	25	"	5,11,8, 26,15,23, 29,30,34, 4,6,9	"	50	"	"
P193	2B0 to 2EQ "	"	31	19	5,11,14, 17,15,23, 36,40,34, 18,37,39	"	45	"	"
P194	2B1 to 2EQ "	"	37	"	5,11,14, 17,15,23, 36,40,34, 18,31,39	"	"	"	"
P195	2B2 to 2EQ "	"	18	"	5,11,14, 17,15,23, 36,40,34, 31,37,39	"	"	"	"
P196	2B3 to 2EQ "	"	39	"	5,11,14, 17,15,23, 36,40,34, 18,31,37	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P197	1A0 to 1F0 Propagation delay time (t_{PLH}), logic mode C_L = 50 pF	24,12,16, 23,15,6	26	2	5,11,34, 30,29,8, 9,4,25,3	5	24	ns	16,17
P198	1A1 to 1F1 "	24,12,16, 23,15,9	30	22	5,11,34, 26,29,8, 6,4,25,3	"	"	"	"
P199	1A2 to 1F2 "	24,12,16, 23,15,4	29	21	5,11,34, 26,30,8, 6,9,25,3	"	"	"	"
P200	1A3 to 1F3 "	24,12,16, 23,15,25	8	27	5,11,34, 26,30,29, 6,9,4,3	"	"	"	"
P201	2A0 to 2F0 "	24,12,16, 23,15,31	14	13	5,11,34, 36,17,40, 37,18,39, 32	"	"	"	"
P202	2A1 to 2F1 "	24,12,16, 23,15,37	36	35	5,11,34, 14,17,40, 31,18,39, 32	"	"	"	"
P203	2A2 to 2F2 "	24,12,16, 23,15,18	17	33	5,11,34, 14,36,40, 31,37,39, 32	"	"	"	"
P204	2A3 to 2F3 "	24,12,16, 23,15,39	40	20	5,11,34, 14,36,17, 31,37,18, 32	"	"	"	"
P205	1A0 to 1F0 Propagation delay time (t_{PLH}), logic mode, C_L = 150 pF	24,12,16, 23,15,6	26	2	5,11,34, 30,29,8, 9,4,25,3	8	28	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P206	1A1 to 1F1 Propagation delay time (t_{PLH}), logic mode, C_L = 150 pF	24,12,16, 23,15,9	30	22	5,11,34, 26,29,8, 6,4,25,3	8	28	ns	16,17
P207	1A2 to 1F2 "	24,12,16, 23,15,4	29	21	5,11,34, 26,30,8, 6,9,25,3	"	"	"	"
P208	1A3 to 1F3 "	24,12,16, 23,15,25	8	27	5,11,34, 26,30,29, 6,9,4,3	"	"	"	"
P209	2A0 to 2F0 "	24,12,16, 23,15,31	14	13	5,11,34, 36,17,40, 37,18,39, 32	"	"	"	"
P210	2A1 to 2F1 "	24,12,16, 23,15,37	36	35	5,11,34, 14,17,40, 31,18,39, 32	"	"	"	"
P211	2A2 to 2F2 "	24,12,16, 23,15,18	17	33	5,11,34, 14,36,40, 31,37,39, 32	"	"	"	"
P212	2A3 to 2F3 "	24,12,16, 23,15,39	40	20	5,11,34, 14,36,17, 31,37,18, 32	"	"	"	"
P213	1B0 to 1F0 Propagation delay time (t_{PLH}), logic mode, C_L = 50 pF	24,12,16, 23,15,26	6	2	5,11,34, 30,29,8, 9,4,25,3	5	24	"	"
P214	1B1 to 1F1 Propagation delay time (t_{PLH}), logic mode, C_L = 50 pF	24,12,16, 23,15,30	9	22	5,11,34, 26,29,8, 6,4,25,3	5	24	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P215	1B2 to 1F2 Propagation delay time (t_{PLH}), logic mode, C_L = 50 pF	24,12,16, 23,15,29	4	21	5,11,34, 26,30,8, 6,9,25,3	5	24	ns	16,17
P216	1B3 to 1F3 "	24,12,16, 23,15,8	25	27	5,11,34, 26,30,29, 6,9,4,3	"	"	"	"
P217	2B0 to 2F0 "	24,12,16, 23,15,14	31	13	5,11,34, 36,17,40, 37,18,39, 32	"	"	"	"
P218	2B1 to 2F1 "	24,12,16, 23,15,36	37	35	5,11,34, 14,17,40, 31,18,39, 32	"	"	"	"
P219	2B2 to 2F2 "	24,12,16, 23,15,17	18	33	5,11,34, 14,36,40, 31,37,39, 32	"	"	"	"
P220	2B3 to 2F3 "	24,12,16, 23,15,40	39	20	5,11,34, 14,36,17, 31,37,18, 32	"	"	"	"
P221	1B0 to 1F0 Propagation delay time (t_{PLH}), logic mode, C_L = 150 pF	24,12,16, 23,15,26	6	2	5,11,34, 30,29,8, 9,4,25,3	8	28	"	"
P222	1B1 to 1F1 "	24,12,16, 23,15,30	9	22	5,11,34, 26,29,8, 6,4,25,3	"	"	"	"
P223	1B2 to 1F2 "	24,12,16, 23,15,29	4	21	5,11,34, 26,30,8, 6,9,25,3	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P224	1B3 to 1F3 Propagation delay time (t_{PLH}), logic mode, C_L = 150 pF	24,12,16, 23,15,8	25	27	5,11,34, 26,30,29, 6,9,4,3	8	28	ns	16,17
P225	2B0 to 2F0 "	24,12,16, 23,15,14	31	13	5,11,34, 36,17,40, 37,18,39 32	"	"	"	"
P226	2B1 to 2F1 "	24,12,16, 23,15,36	37	35	5,11,34, 14,17,40, 31,18,39 32	"	"	"	"
P227	2B2 to 2F2 "	24,12,16, 23,15,17	18	33	5,11,34, 14,36,40, 31,37,39 32	"	"	"	"
P228	2B3 to 2F3 "	24,12,16, 23,15,40	39	20	5,11,34, 14,36,17, 31,37,18 32	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1A0 to 1F0 Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	5,34,30,29, 11,3,8,6,32	26	2	24,12,9,4, 16,23,25, 15	5	21	ns	16,17
Q2	1A1 to 1F1 "	5,34,26,29, 11,3,8,9,32	30	22	24,12,6,4, 16,23,25, 15	"	"	"	"
Q3	1A2 to 1F2 "	5,34,26,30, 11,3,8,4,32	29	21	24,12,6,9, 16,23,25, 15	"	"	"	"
Q4	1A3 to 1F3 "	5,34,26,30, 11,3,29,25, 32	8	27	24,12,6,9, 16,23,4, 15	"	"	"	"
Q5	2A0 to 2F0 "	5,34,36,17, 11,3,40,31, 32	14	13	24,12,37, 18,16,23, 39,15	"	"	"	"
Q6	2A1 to 2F1 "	5,34,14,17, 11,3,40,37, 32	36	35	24,12,31, 18,16,23, 39,15	"	"	"	"
Q7	2A2 to 2F2 "	5,34,14,36, 11,3,40,18, 32	17	33	24,12,31, 37,16,23, 39,15	"	"	"	"
Q8	2A3 to 2F3 "	5,34,14,36, 11,3,17,39, 32	40	20	24,12,31, 37,16,23, 18,15	"	"	"	"
Q9	1A0 to 1F0 Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	5,34,30,29, 11,3,8,6, 32	26	2	24,12,9, 4,16,23, 25,15	8	24	"	"
Q10	1A1 to 1F1 Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	5,34,26,29, 11,3,8,9, 32	30	22	24,12,6, 4,16,23, 25,15	8	24	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q11	1A2 to 1F2 Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	5,34,26,30, 11,3,8,4,32	29	21	24,12,6,9, 16,23,25, 15	8	24	ns	16,17
Q12	1A3 to 1F3 "	5,34,26,30, 11,3,29,25, 32	8	27	24,12,6,9, 16,23,4, 15	"	"	"	"
Q13	2A0 to 2F0 "	5,34,36, 17,11,3, 40,31,32	14	13	24,12,37, 18,16,23, 39,15	"	"	"	"
Q14	2A1 to 2F1 "	5,34,14, 17,11,3, 40,37,32	36	35	24,12,31, 18,16,23, 39,15	"	"	"	"
Q15	2A2 to 2F2 "	5,34,14, 36,11,3, 40,18,32	17	33	24,12,31, 37,16,23, 39,15	"	"	"	"
Q16	2A3 to 2F3 "	5,34,14, 36,11,3, 17,39,32	40	20	24,12,31, 37,16,23, 18,15	"	"	"	"
Q17	1B0 to 1F0 Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	5,34,11, 3,32	6	2	24,12,8, 29,16,23, 30,26,15	5	21	"	"
Q18	1B1 to 1F1 "	"	9	22	"	"	"	"	"
Q19	1B2 to 1F2 "	"	4	21	"	"	"	"	"
Q20	1B3 to 1F3 "	"	25	27	"	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q21	2B0 to 2F0 Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	5,34,11, 3,32	31	13	24,12,40, 17,16,23, 36,14,15	5	21	ns	16,17
Q22	2B1 to 2F1 "	"	37	35	"	"	"	"	"
Q23	2B2 to 2F2 "	"	18	33	"	"	"	"	"
Q24	2B3 to 2F3 "	"	39	20	"	"	"	"	"
Q25	1B0 to 1F0 Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	"	6	2	24,12,8, 29,16,23, 30,26,15	8	24	"	"
Q26	1B1 to 1F1 "	"	9	22	"	"	"	"	"
Q27	1B2 to 1F2 "	"	4	21	"	"	"	"	"
Q28	1B3 to 1F3 "	"	25	27	"	"	"	"	"
Q29	2B0 to 2F0 "	"	31	13	24,12,40, 17,16,23, 36,14,15	"	"	"	"
Q30	2B1 to 2F1 "	"	37	35	"	"	"	"	"
Q31	2B2 to 2F2 "	"	18	33	"	"	"	"	"
Q32	2B3 to 2F3 "	"	39	20	"	"	"	"	"
Q33	1CN to 1F0 Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36	3	2	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	3	17	"	"
Q34	1CN to 1F1 Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36	3	22	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	3	17	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q35	1CN to 1F2 Propagation delay time (t_{PHL}), sum mode, C_L = 50 pF	5,8,11,14, 17,26,29, 30,34,36	3	21	4,6,9,12, 15,16,18, 23,24,25 31,37,39	3	17	ns	16,17
Q36	1CN to 1F3 "	"	3	27	"	"	"	"	"
Q37	2CN to 2F0 "	"	32	13	"	"	"	"	"
Q38	2CN to 2F1 "	"	"	35	"	"	"	"	"
Q39	2CN to 2F2 "	"	"	33	"	"	"	"	"
Q40	2CN to 2F3 "	"	"	20	"	"	"	"	"
Q41	1CN to 1F0 Propagation delay time (t_{PHL}), sum mode, C_L = 150 pF	"	3	2	"	5	20	"	"
Q42	1CN to 1F1 "	"	"	22	"	"	"	"	"
Q43	1CN to 1F2 "	"	"	21	"	"	"	"	"
Q44	1CN to 1F3 "	"	"	27	"	"	"	"	"
Q45	2CN to 2F0 "	"	32	13	"	"	"	"	"
Q46	2CN to 2F1 "	"	"	35	"	"	"	"	"
Q47	2CN to 2F2 "	"	"	33	"	"	"	"	"
Q48	2CN to 2F3 "	"	"	20	"	"	"	"	"
Q49	1CN to 1C0 Propagation delay time (t_{PHL}), sum mode C_L = 50 pF	"	3	28	"	3	14	"	"
Q50	2CN to 2C0 Propagation delay time (t_{PHL}), sum mode C_L = 50 pF	"	32	38	"	3	14	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q51	1CN to 1CO Propagation delay time (t_{PHL}), sum mode C_L = 150 pF	5,8,11,14, 17,26,29, 30,34,36	3	28	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	5	18	ns	16,17
Q52	2CN to 2CO Propagation delay time (t_{PHL}), sum mode C_L = 150 pF	5,8,11,14, 17,26,29, 30,34,36	32	38	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	"	18	"	"
Q53	1AO to 1CO Propagation delay time (t_{PHL}), sum mode C_L = 50 pF	5,34,11, 9,4,25	26	28	24,12,16, 23,15,6, 30,29,8, 3	"	20	"	"
Q54	1A1 to 1CO "	5,34,11, 6,4,25	30	"	24,12,16, 23,15,9, 26,29,8, 3	"	"	"	"
Q55	1A2 to 1CO "	5,34,11, 6,9,25	29	"	24,12,16, 23,15,4, 26,30,8, 3	"	"	"	"
Q56	1A3 to 1CO "	5,34,11, 6,9,4	8	"	24,12,16, 23,15,25, 26,30,29, 3	"	"	"	"
Q57	2AO to 2CO "	5,34,11, 37,18,39	14	38	24,12,16, 23,15,31, 36,17,40, 32	"	"	"	"
Q58	2A1 to 2CO "	5,34,11, 31,18,39	36	38	24,12,16, 23,15,37, 14,17,40, 32	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q59	2A2 to 2C0 Propagation delay time (t_{PHL}), sum mode C_L = 50 pF	5,34,11, 31,37,39	17	38	24,12,16, 23,15,18, 14,36,40, 32	5	20	ns	16,17
Q60	2A3 to 2C0 Propagation delay time (t_{PHL}), sum mode C_L = 50 pF	5,34,11, 31,37,18	40	38	24,12,16, 23,15,39, 14,36,17, 32	5	20	"	"
Q61	1A0 to 1C0 Propagation delay time (t_{PHL}), sum mode C_L = 150 pF	5,34,11, 9,4,25	26	28	24,12,16, 23,15,6, 30,29,8, 3	8	24	"	"
Q62	1A1 to 1C0 "	5,34,11, 6,4,25	30	"	24,12,16, 23,15,9, 26,29,8, 3	"	"	"	"
Q63	1A2 to 1C0 "	5,34,11, 6,9,25	29	"	24,12,16, 23,15,4, 26,30,8, 3	"	"	"	"
Q64	1A3 to 1C0 "	5,34,11, 6,9,4	8	"	24,12,16, 23,15,25, 26,30,29, 3	"	"	"	"
Q65	2A0 to 2C0 "	5,34,11, 37,18,39	14	38	24,12,16, 23,15,31, 36,17,40, 32	"	"	"	"
Q66	2A1 to 2C0 "	5,34,11, 31,18,39	36	38	24,12,16, 23,15,37, 14,17,40, 32	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q67	2A2 to 2C0 Propagation delay time (t_{PHL}), sum mode C_L = 150 pF	5,34,11, 31,37,39	17	38	24,12,16, 23,15,18, 14,36,40, 32	8	24	ns	16,17
Q68	2A3 to 2C0 Propagation delay time (t_{PHL}), sum mode C_L = 150 pF	5,34,11, 31,37,18	40	38	24,12,16, 23,15,39, 14,36,17, 32	8	24	"	"
Q69	1B0 to 1C0 Propagation delay time (t_{PHL}), sum mode C_L = 50 pF	5,34,11, 9,4,25	6	28	24,12,16, 23,15,26, 30,29,8, 3	5	20	"	"
Q70	1B1 to 1C0 "	5,34,11, 6,4,25	9	"	"	"	"	"	"
Q71	1B2 to 1C0 "	5,34,11, 6,9,25	4	"	"	"	"	"	"
Q72	1B3 to 1C0 "	5,34,11, 6,9,4	25	"	"	"	"	"	"
Q73	2B0 to 2C0 "	5,34,11, 37,18,39	31	38	24,12,16 23,15,14, 36,17,40, 32	"	"	"	"
Q74	2B1 to 2C0 "	5,34,11, 31,18,39	37	"	"	"	"	"	"
Q75	2B2 to 2C0 "	5,34,11, 31,37,39	18	"	"	"	"	"	"
Q76	2B3 to 2C0 "	5,34,11, 31,37,18	39	"	"	"	"	"	"
Q77	1B0 to 1C0 Propagation delay time (t_{PHL}), sum mode C_L = 150 pF	5,34,11, 9,4,25	6	28	24,12,16, 23,15,26, 30,29,8,3	8	24	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q78	1B1 to 1C0 Propagation delay time (t_{PHL}), sum mode C_L = 150 pF	5,34,11, 6,4,25	9	28	24,12,16, 23,15,26, 30,29,8,3	8	24	ns	16,17
Q79	1B2 to 1C0 "	5,34,11, 6,9,25	4	28	"	"	"	"	"
Q80	1B3 to 1C0 "	5,34,11, 6,9,4	25	28	"	"	"	"	"
Q81	2B0 to 2C0 "	5,34,11, 37,18,39	31	38	24,12,16, 23,15,14, 36,17,40, 32	"	"	"	"
Q82	2B1 to 2C0 "	5,34,11, 31,18,39	37	"	"	"	"	"	"
Q83	2B2 to 2C0 "	5,34,11, 31,37,39	18	"	"	"	"	"	"
Q84	2B3 to 2C0 "	5,34,11, 31,37,18	39	"	"	"	"	"	"
Q85	1A0 to 1F0 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12,16, 30,29,8	26	2	5,34,11, 23,15,6, 9,4,25,3	5	26	"	"
Q86	1A1 to 1F1 "	24,12,16, 26,29,8	30	22	"	"	"	"	"
Q87	1A2 to 1F2 "	24,12,16, 26,30,8	29	21	"	"	"	"	"
Q88	1A3 to 1F3 "	24,12,16, 26,30,29	8	27	"	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q89	2A0 to 2F0 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12,16, 36,17,40	14	13	5,34,11, 23,15,31, 37,18,39, 32	5	26	ns	16,17
Q90	2A1 to 2F1 "	24,12,16, 14,17,40	36	35	"	"	"	"	"
Q91	2A2 to 2F2 "	24,12,16, 14,36,40	17	33	"	"	"	"	"
Q92	2A3 to 2F3 "	24,12,16, 14,36,17	40	20	"	"	"	"	"
Q93	1A0 to 1F0 Propagation delay time (t_{PHL}), difference mode C_L = 150 pF	24,12,16, 30,29,8	26	2	5,34,11, 23,15,6, 9,4,25,3	8	30	"	"
Q94	1A1 to 1F1 "	24,12,16, 26,29,8	30	22	"	"	"	"	"
Q95	1A2 to 1F2 "	24,12,16, 26,30,8	29	21	"	"	"	"	"
Q96	1A3 to 1F3 "	24,12,16, 26,30,29	8	27	"	"	"	"	"
Q97	2A0 to 2F0 "	24,12,16, 36,17,40	14	13	5,34,11, 23,15,31, 37,18,39, 32	"	"	"	"
Q98	2A1 to 2F1 "	24,12,16, 14,17,40	36	35	5,34,11, 23,15,31, 37,18,39, 32	"	"	"	"
Q99	2A2 to 2F2 "	24,12,16, 14,36,40	17	33	5,34,11, 23,15,31, 37,18,39, 32	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q100	2A3 to 2F3 Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	24,12,16, 14,36,17	40	20	5,34,11, 23,15,31, 37,18,39, 32	8	30	ns	16,17
Q101	1B0 to 1F0 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12,16, 26,30,29 8	6	2	5,34,11, 23,15,9, 4,25,3	5	26	"	"
Q102	1B1 to 1F1 "	"	9	22	5,34,11, 23,15,6, 4,25,3	"	"	"	"
Q103	1B2 to 1F2 "	"	4	21	5,34,11, 23,15,6, 9,25,3	"	"	"	"
Q104	1B3 to 1F3 "	"	25	27	5,34,11, 23,15,6, 9,4,3	"	"	"	"
Q105	2B0 to 2F0 "	24,12,16, 14,36,17, 40	31	13	5,34,11, 23,15,37, 18,39,32	"	"	"	"
Q106	2B1 to 2F1 "	"	37	35	5,34,11, 23,15,31, 18,39,32	"	"	"	"
Q107	2B2 to 2F2 "	"	18	33	5,34,11, 23,15,31, 37,39,32	"	"	"	"
Q108	2B3 to 2F3 "	"	39	20	5,34,11, 23,15,31, 37,18,32	"	"	"	"
Q109	1B0 to 1F0 Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	24,12,16, 26,30,29, 8	6	2	5,34,11, 23,15,9, 4,25,3	8	30	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q110	1B1 to 1F1 Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	24,12,16, 26,30,29, 8	9	22	5,34,11, 23,15,6, 4,25,3	8	30	ns	16,17
Q111	1B2 to 1F2 "	"	4	21	5,34,11, 23,15,6, 9,25,3	"	"	"	"
Q112	1B3 to 1F3 "	"	25	27	5,34,11, 23,15,6, 9,4,3	"	"	"	"
Q113	2B0 to 2F0 "	24,12,16, 14,36,17, 40	31	13	5,34,11, 23,15,37, 18,39,32	"	"	"	"
Q114	2B1 to 2F1 "	"	37	35	5,34,11, 23,15,31, 18,39,32	"	"	"	"
Q115	2B2 to 2F2 "	"	18	33	5,34,11, 23,15,31, 37,39,32	"	"	"	"
Q116	2B3 to 2F3 "	"	39	20	5,34,11, 23,15,31, 37,18,32	"	"	"	"
Q117	1CN to 1F0 Propagation delay time (t_{PHL}), difference mode C_L = 50 pF	24,12,16, 26,30,29, 8,6,9,4, 25	3	2	5,34,11, 23,15	13	17	"	"
Q118	1CN to 1F1 "	"	"	22	"	"	"	"	"
Q119	1CN to 1F2 "	"	"	21	"	"	"	"	"
Q120	1CN to 1F3 "	"	"	27	"	"	"	"	"
Q121	2CN to 2F0 "	24,12,16, 14,36,17, 40,31,37, 18,39	32	13	"	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q122	2CN to 2F1 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12,16, 14,36,17, 40,31,37, 18,39	32	35	5,34,11, 23,15	3	17	ns	16,17
Q123	2CN to 2F2 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12,16, 14,36,17, 40,31,37, 18,39	32	33	"	3	17	"	"
Q124	2CN to 2F3 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12,16, 14,36,17, 40,31,37, 18,39	32	20	"	3	17	"	"
Q125	1CN to 1F0 Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	24,12,16, 26,30,29, 8,6,9,4, 25	3	2	"	5	20	"	"
Q126	1CN to 1F1 "	"	"	22	"	"	"	"	"
Q127	1CN to 1F2 "	"	"	21	"	"	"	"	"
Q128	1CN to 1F3 "	"	"	27	"	"	"	"	"
Q129	2CN to 2F0 "	24,12,16, 14,36,17, 40,31,37, 18,39	32	13	"	"	"	"	"
Q130	2CN to 2F1 "	"	"	35	"	"	"	"	"
Q131	2CN to 2F2 "	"	"	33	"	"	"	"	"
Q132	2CN to 2F3 "	"	"	20	"	"	"	"	"
Q133	1A0 to 1C0 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12, 16,6	26	28	5,34,11, 23,15,30, 29,8,9,4, 25,3	"	27	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q134	1A1 to 1C0 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12, 16,9	30	28	5,34,11, 23,15,26, 29,8,6,4, 25,3	5	27	ns	16,17
Q135	1A2 to 1C0 "	24,12, 16,4	29	28	5,34,11, 23,15,26, 30,8,6,9, 25,3	"	"	"	"
Q136	1A3 to 1C0 "	24,12, 16,25	8	28	5,34,11, 23,15,26, 30,29,6, 9,4,3	"	"	"	"
Q137	2A0 to 2C0 "	24,12, 16,31	14	38	5,34,11, 23,15,36, 17,40,37, 18,39,32	"	"	"	"
Q138	2A1 to 2C0 "	24,12, 16,37	36	"	5,34,11, 23,15,14, 17,40,31, 18,39,32	"	"	"	"
Q139	2A2 to 2C0 "	24,12, 16,18	17	"	5,34,11, 23,15,14, 36,40,31, 37,39,32	"	"	"	"
Q140	2A3 to 2C0 "	24,12, 16,39	40	"	5,34,11, 23,15,14, 36,17,31, 37,18,32	"	"	"	"
Q141	1A0 to 1C0 Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	24,12, 16,6	26	28	5,34,11, 23,15,30, 29,8,9,4, 25,3	8	31	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q142	1A1 to 1CO Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	24,12, 16,9	30	28	5,34,11, 23,15,26, 29,8,6,4, 25,3	8	31	ns	16,17
Q143	1A2 to 1CO "	24,12, 16,4	29	28	5,34,11, 23,15,26, 30,8,6,9, 25,3	"	"	"	"
Q144	1A3 to 1CO "	24,12, 16,25	8	28	5,34,11, 23,15,26, 30,29,6, 9,4,3	"	"	"	"
Q145	2A0 to 2CO "	24,12, 16,31	14	38	5,34,11, 23,15,36, 17,40,37, 18,39,32	"	"	"	"
Q146	2A1 to 2CO "	24,12, 16,37	36	"	5,34,11, 23,15,14, 17,40,31, 18,39,32	"	"	"	"
Q147	2A2 to 2CO "	24,12, 16,18	17	"	5,34,11, 23,15,14, 36,40,31, 37,39,32	"	"	"	"
Q148	2A3 to 2CO "	24,12, 16,39	40	"	5,34,11, 23,15,14, 36,17,31, 37,18,32	"	"	"	"
Q149	1B0 to 1CO Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12, 16	6	28	5,34,11, 23,15,26, 30,29,8, 9,4,25,3	5	27	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q150	1B1 to 1C0 Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	24,12, 16	9	28	5,34,11, 23,15,26, 30,29,8, 6,4,25,3	5	27	ns	16,17
Q151	1B2 to 1C0 "	"	4	28	5,34,11, 23,15,26, 30,29,8, 6,9,25,3	"	"	"	"
Q152	1B3 to 1C0 "	"	25	28	5,34,11, 23,15,26, 30,29,8, 6,9,4,3	"	"	"	"
Q153	2B0 to 2C0 "	"	31	38	5,34,11, 23,15,14, 36,17,40, 37,18,39, 32	"	"	"	"
Q154	2B1 to 2C0 "	"	37	"	5,34,11, 23,15,14, 36,17,40, 31,18,39, 32	"	"	"	"
Q155	2B2 to 2C0 "	"	18	"	5,34,11, 23,15,14, 36,17,40, 31,37,39, 32	"	"	"	"
Q156	2B3 to 2C0 "	"	39	"	5,34,11, 23,15,14, 36,17,40, 31,37,18, 32	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q157	1B0 to 1C0 Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	24,12, 16	6	28	5,34,11, 23,15,26, 30,29,8, 9,4,25,3	8	31	ns	16,17
Q158	1B1 to 1C0 "	"	9	"	5,34,11, 23,15,26, 30,29,8, 6,4,25,3	"	"	"	"
Q159	1B2 to 1C0 "	"	4	"	5,34,11, 23,15,26, 30,29,8, 6,9,25,3	"	"	"	"
Q160	1B3 to 1C0 "	"	25	"	5,34,11, 23,15,26, 30,29,8, 6,9,4,3	"	"	"	"
Q161	2B0 to 2C0 "	"	31	38	5,34,11, 23,15,14, 36,17,40, 37,18,39, 32	"	"	"	"
Q162	2B1 to 2C0 "	"	37	"	5,34,11, 23,15,14, 36,17,40, 31,18,39, 32	"	"	"	"
Q163	2B2 to 2C0 "	"	18	"	5,34,11, 23,15,14, 36,17,40, 31,37,39, 32	"	"	"	"
Q164	2B3 to 2C0 "	"	39	"	5,34,11, 23,15,14, 36,17,40, 31,37,18, 32	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q165	1A0 to 1EQ Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	3,12,4,6, 9,16,24, 25,8,32, 29,30	26	7	5,11,15, 23,34	10	33	ns	16,17
Q166	1A1 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,29	30	"	"	"	"	"	"
Q167	1A2 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,30	29	"	"	"	"	"	"
Q168	1A3 to 1EQ "	3,12,4,6, 9,16,24, 25,26,32, 29,30	8	"	"	"	"	"	"
Q169	2A0 to 2EQ "	3,12,18, 31,16,24 37,39,32, 17,36,40	14	19	"	"	"	"	"
Q170	2A1 to 2EQ "	3,12,18, 31,16,24 37,39,32, 14,17,40	36	"	"	"	"	"	"
Q171	2A2 to 2EQ "	3,12,18, 31,16,24 37,39,32, 14,36,40	17	"	"	"	"	"	"
Q172	2A3 to 2EQ "	3,12,18, 31,16,24 37,39,32, 14,17,36	40	"	"	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q173	1A0 to 1EQ Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	3,12,4,6, 9,16,24, 25,8,32, 29,30	26	7	5,11,15, 23,34	10	37	ns	16,17
Q174	1A1 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,29	30	"	"	"	"	"	"
Q175	1A2 to 1EQ "	3,12,4,6, 9,16,24, 25,8,32, 26,30	29	"	"	"	"	"	"
Q176	1A3 to 1EQ "	3,12,4,6, 9,16,24, 25,26,32, 29,30	8	"	"	"	"	"	"
Q177	2A0 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 17,36,40	14	19	"	"	"	"	"
Q178	2A1 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 14,17,40	36	"	"	"	"	"	"
Q179	2A2 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 14,36,40	17	"	"	"	"	"	"
Q180	2A3 to 2EQ "	3,12,18, 31,16,24, 37,39,32, 14,17,36	40	"	"	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q181	1BO to 1EQ Propagation delay time (t_{PHL}), difference mode, C_L = 50 pF	3,12,16, 24,32	6	7	5,11,8, 26,15,23, 29,30,34, 4,9,25	10	33	ns	16,17
Q182	1B1 to 1EQ "	"	9	"	5,11,8, 26,15,23, 29,30,34, 4,6,25	"	"	"	"
Q183	1B2 to 1EQ "	"	4	"	5,11,8, 26,15,23, 29,30,34, 6,9,25	"	"	"	"
Q184	1B3 to 1EQ "	"	25	"	5,11,8, 26,15,23, 29,30,34, 4,6,9	"	"	"	"
Q185	2BO to 2EQ "	"	31	19	5,11,14, 17,15,23, 36,40,34, 18,37,39	"	"	"	"
Q186	2B1 to 2EQ "	"	37	"	5,11,14, 17,15,23, 36,40,34, 18,31,39	"	"	"	"
Q187	2B2 to 2EQ "	"	18	"	5,11,14, 17,15,23, 36,40,34, 31,37,39	"	"	"	"
Q188	2B3 to 2EQ "	"	39	"	5,11,14, 17,15,23, 36,40,34, 18,31,37	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q189	1B0 to 1EQ Propagation delay time (t_{PHL}), difference mode, C_L = 150 pF	3,12,16, 24,32	6	7	5,11,8, 26,15,23, 29,30,34, 4,9,25	10	37	ns	16,17
Q190	1B1 to 1EQ "	"	9	"	5,11,8, 26,15,23, 29,30,34, 4,6,25	"	"	"	"
Q191	1B2 to 1EQ "	"	4	"	5,11,8, 26,15,23, 29,30,34, 6,9,25	"	"	"	"
Q192	1B3 to 1EQ "	"	25	"	5,11,8, 26,15,23, 29,30,34, 4,6,9	"	"	"	"
Q193	2B0 to 2EQ "	"	31	19	5,11,14, 17,15,23, 36,40,34, 18,37,39	"	"	"	"
Q194	2B1 to 2EQ "	"	37	"	5,11,14, 17,15,23, 36,40,34, 18,31,39	"	"	"	"
Q195	2B2 to 2EQ "	"	18	"	5,11,14, 17,15,23, 36,40,34, 31,37,39	"	"	"	"
Q196	2B3 to 2EQ "	"	39	"	5,11,14, 17,15,23, 36,40,34, 18,31,37	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q197	1A0 to 1F0 Propagation delay time (t_{PHL}), logic mode C_L = 50 pF	24,12,16, 23,15,6	26	2	5,11,34, 30,29,8, 9,4,25,3	5	26	ns	16,17
Q198	1A1 to 1F1 "	24,12,16, 23,15,9	30	22	5,11,34, 26,29,8, 6,4,25,3	"	"	"	"
Q199	1A2 to 1F2 "	24,12,16, 23,15,4	29	21	5,11,34, 26,30,8, 6,9,25,3	"	"	"	"
Q200	1A3 to 1F3 "	24,12,16, 23,15,25	8	27	5,11,34, 26,30,29, 6,9,4,3	"	"	"	"
Q201	2A0 to 2F0 "	24,12,16, 23,15,31	14	13	5,11,34, 36,17,40, 37,18,39, 32	"	"	"	"
Q202	2A1 to 2F1 "	24,12,16, 23,15,37	36	35	5,11,34, 14,17,40, 31,18,39, 32	"	"	"	"
Q203	2A2 to 2F2 "	24,12,16, 23,15,18	17	33	5,11,34, 14,36,40, 31,37,39, 32	"	"	"	"
Q204	2A3 to 2F3 "	24,12,16, 23,15,39	40	20	5,11,34, 14,36,17, 31,37,18, 32	"	"	"	"
Q205	1A0 to 1F0 Propagation delay time (t_{PHL}), logic mode, C_L = 150 pF	24,12,16, 23,15,6	26	2	5,11,34, 30,29,8, 9,4,25,3	8	30	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q206	1A1 to 1F1 Propagation delay time (t_{PHL}), logic mode, C_L = 150 pF	24,12,16, 23,15,9	30	22	5,11,34, 26,29,8, 6,4,25,3	8	30	ns	16,17
Q207	1A2 to 1F2 "	24,12,16, 23,15,4	29	21	5,11,34, 26,30,8, 6,9,25,3	"	"	"	"
Q208	1A3 to 1F3 "	24,12,16, 23,15,25	8	27	5,11,34, 26,30,29, 6,9,4,3	"	"	"	"
Q209	2A0 to 2F0 "	24,12,16, 23,15,31	14	13	5,11,34, 36,17,40, 37,18,39, 32	"	"	"	"
Q210	2A1 to 2F1 "	24,12,16, 23,15,37	36	35	5,11,34, 14,17,40, 31,18,39, 32	"	"	"	"
Q211	2A2 to 2F2 "	24,12,16, 23,15,18	17	33	5,11,34, 14,36,40, 31,37,39, 32	"	"	"	"
Q212	2A3 to 2F3 "	24,12,16, 23,15,39	40	20	5,11,34, 14,36,17, 31,37,18, 32	"	"	"	"
Q213	1B0 to 1F0 Propagation delay time (t_{PHL}), logic mode, C_L = 50 pF	24,12,16, 23,15,26	6	2	5,11,34, 30,29,8, 9,4,25,3	5	26	"	"
Q214	1B1 to 1F1 Propagation delay time (t_{PHL}), logic mode, C_L = 50 pF	24,12,16, 23,15,30	9	22	5,11,34, 26,29,8, 6,4,25,3	5	26	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

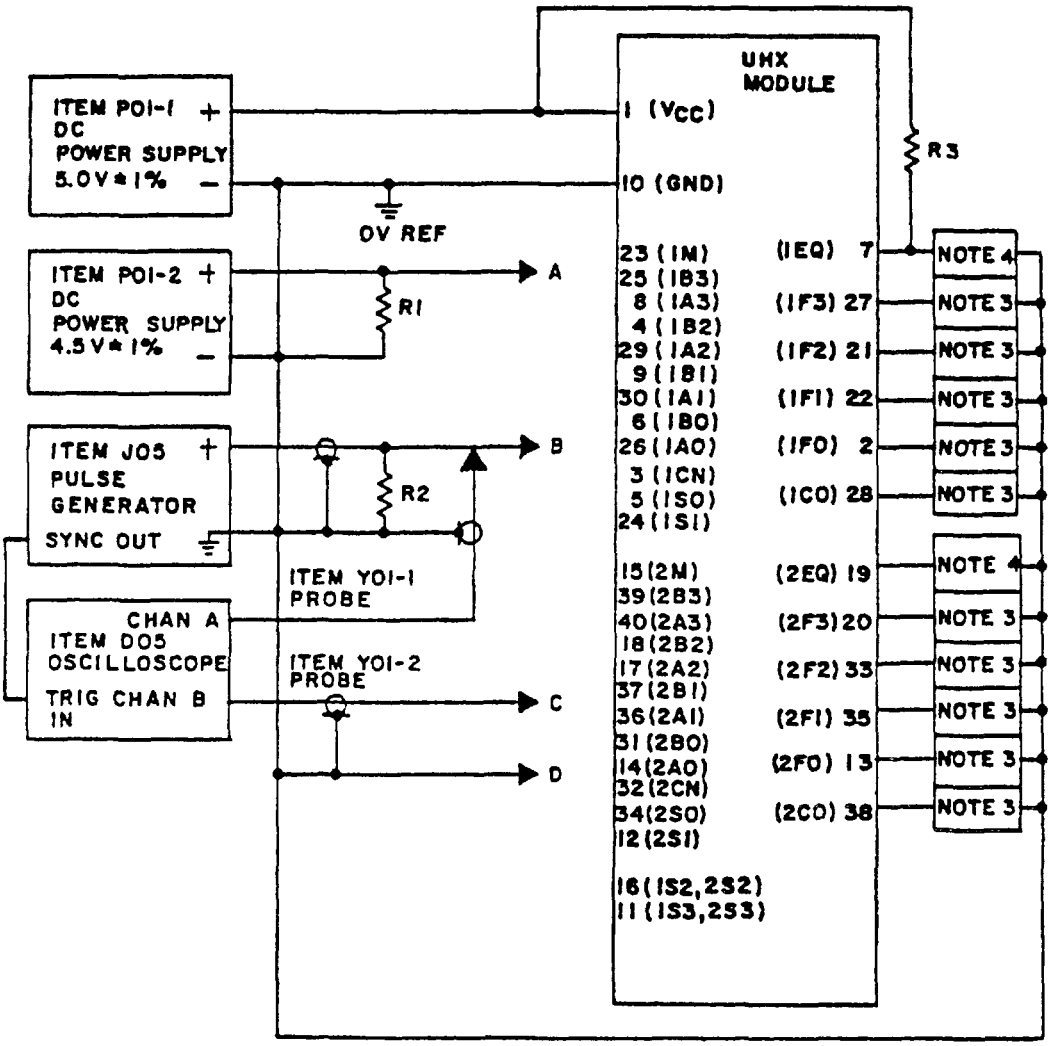
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q215	1B2 to 1F2 Propagation delay time (t_{PHL}), logic mode, C_L = 50 pF	24,12,16, 23,15,29	4	21	5,11,34, 26,30,8, 6,9,25,3	5	26	ns	16,17
Q216	1B3 to 1F3 "	24,12,16, 23,15,8	25	27	5,11,34, 26,30,29, 6,9,4,3	"	"	"	"
Q217	2B0 to 2F0 "	24,12,16, 23,15,14	31	13	5,11,34, 36,17,40, 37,18,39, 32	"	"	"	"
Q218	2B1 to 2F1 "	24,12,16, 23,15,36	37	35	5,11,34, 14,17,40, 31,18,39, 32	"	"	"	"
Q219	2B2 to 2F2 "	24,12,16, 23,15,17	18	33	5,11,34, 14,36,40, 31,37,39, 32	"	"	"	"
Q220	2B3 to 2F3 "	24,12,16, 23,15,40	39	20	5,11,34, 14,36,17, 31,37,18, 32	"	"	"	"
Q221	1B0 to 1F0 Propagation delay time (t_{PHL}), logic mode, C_L = 150 pF	24,12,16, 23,15,26	6	2	5,11,34, 30,29,8, 9,4,25,3	8	30	"	"
Q222	1B1 to 1F1 "	24,12,16, 23,15,30	9	22	5,11,34, 26,29,8, 6,4,25,3	8	30	"	"
Q223	1B2 to 1F2 "	24,12,16, 23,15,29	4	21	5,11,34, 26,30,8, 6,9,25,3	8	30	"	"

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TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q224	1B3 to 1F3 Propagation delay time (t_{PHL}), logic mode, C_L = 150 pF	24,12,16, 23,15,8	25	27	5,11,34, 26,30,29, 6,9,4,3	8	30	ns	16,17
Q225	2B0 to 2F0 "	24,12,16, 23,15,14	31	13	5,11,34, 36,17,40, 37,18,39 32	"	"	"	"
Q226	2B1 to 2F1 "	24,12,16, 23,15,36	37	35	5,11,34, 14,17,40, 31,18,39 32	"	"	"	"
Q227	2B2 to 2F2 "	24,12,16, 23,15,17	18	33	5,11,34, 14,36,40, 31,37,39 32	"	"	"	"
Q228	2B3 to 2F3 "	24,12,16, 23,15,40	39	20	5,11,34, 14,36,17, 31,37,18 32	"	"	"	"

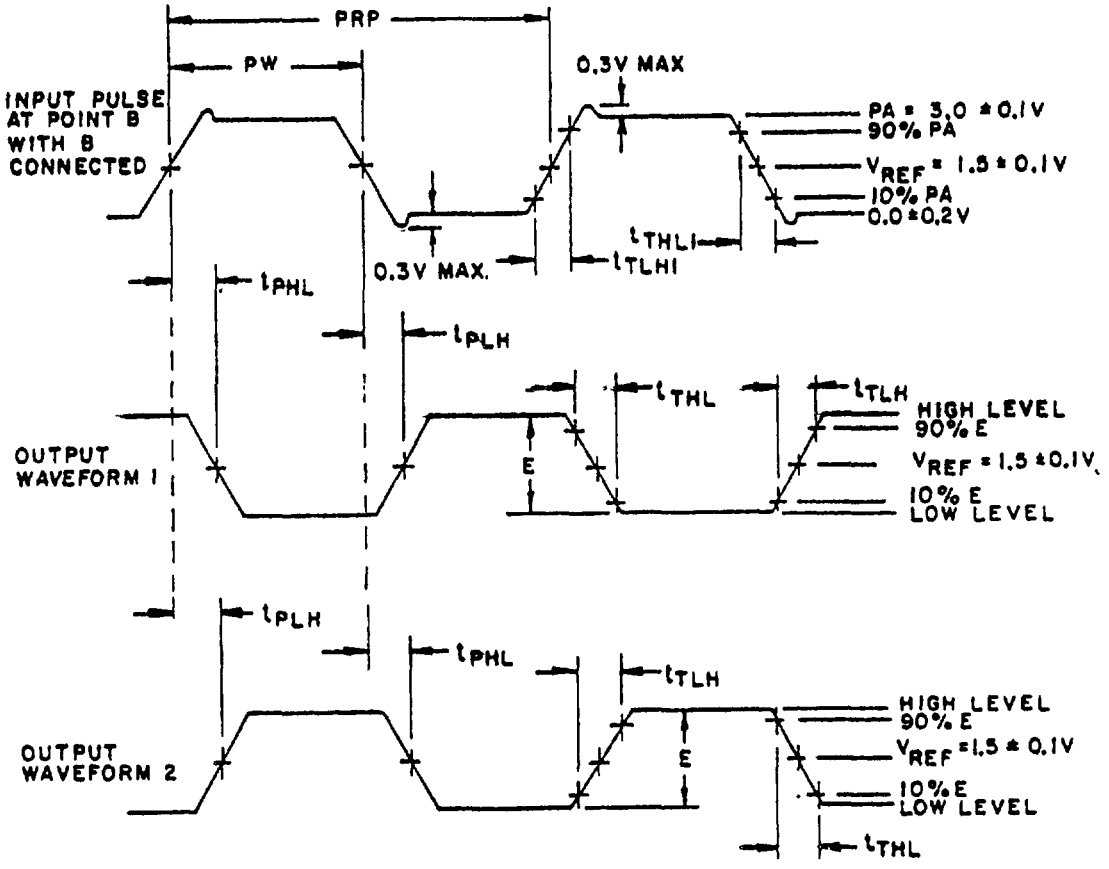
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NOTES.
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 1.21 kΩ ±1%, 1/4 W.
3. Use load circuit 1, figure 12.
4. Use load circuit 2, figure 12.
5. Only the output under test needs to be loaded.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV, XVI, and XVII.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLHI} = t_{THLI}$ = 4 ±1 ns.
2. For test conditions M21-M24 (EQ outputs), t_{TLH} shall be measured from 10% E to 50% E.
3. For test conditions N21-N24 (EQ outputs), t_{THL} shall be measured from 50% E to 10% E.
4. All voltage measurements are referenced to ground.

FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV, XVI and XVII.

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TABLE XVIII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1F0 Output crosstalk, low level	5,3,11,14, 17,34,36, 40	8,29,30,32	2	4,6,9,12, 26,15,16, 18,23,24, 25,31,37, 39	0	900	mV	18,19
R2	1F1 "	"	8,29,26,32	22	4,6,9,12, 15,16,18, 23,24,25, 31,37,30, 39	"	"	"	"
R3	1F2 "	"	8,26,30,32	21	4,6,9,12, 15,16,18, 29,23,24, 25,31,37, 39	"	"	"	"
R4	1F3 "	"	26,29,30,32	27	4,6,9,12, 15,16,18, 8,23,24, 25,31,37, 39	"	"	"	"
R5	1EQ "	"	26,29,30,32	7	4,6,9,12, 15,16,18, 8,23,24, 25,31,37, 39	"	"	"	"
R6	2F0 "	5,8,11,26, 29,30,32, 34	3,17,36,40	13	4,6,9,12, 15,16,18, 14,23,24, 25,31,37, 39	"	"	"	"
R7	2F1 "	5,8,11,26, 29,30,32, 34	3,14,17,40	35	4,6,9,12, 15,16,18, 23,24,25, 31,37,39, 36	"	"	"	"

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TABLE XVIII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R8	2F2 Output crosstalk, low level	5,8,11,26, 29,30,32, 34	3,14,36,40	33	4,6,9,12, 15,16,18, 17,23,24, 25,31,37, 39	0	900	mV	18,19
R9	2F3 "	5,8,11,26, 29,30,32, 34	3,14,17,36	20	4,6,9,12, 15,16,18, 40,23,24, 25,31,37, 39	"	"	"	"
R10	2EQ "	5,8,11,26, 29,30,32, 34	3,14,17,36	19	4,6,9,12, 40,15,16, 18,23,24, 25,31,37, 39	"	"	"	"
R11	1C0 "	5,8,11,23, 14,17,25, 26,29,30, 34,36,40	4,6,9,32	28	3,12,15, 16,18,24, 31,37,39	"	"	"	"
R12	2C0 "	5,8,11,15, 14,17,39, 26,29,30, 34,36,40	3,18,31,37	38	4,6,9,12, 23,16,24, 25,32	"	"	"	"
R13	1F0 Output crosstalk, delayed, low level	5,3,11,14, 17,34,36, 40	8,29,30,32	2	4,6,9,12, 26,15,16, 18,23,24, 25,31,37, 39	"	500	"	"
R14	1F1 Output crosstalk, delayed, low level	5,3,11,14, 17,34,36, 40	8,29,26,32	22	4,6,9,12, 15,16,18, 23,24,25, 31,37,30, 39	"	500	"	"

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TABLE XVIII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R15	1F2 Output crosstalk, delayed, low level	5,3,11,14, 17,34,36, 40	8,26,30,32	21	4,6,9,12, 15,16,18, 29,23,24, 25,31,37, 39	0	500	mV	18,19
R16	1F3 "	"	26,29,30,32	27	4,6,9,12, 15,16,18, 8,23,24, 25,31,37, 39	"	"	"	"
R17	1EQ "	"	26,29,30,32	7	4,6,9,12, 15,16,18, 8,23,24, 25,31,37, 39	"	"	"	"
R18	2F0 "	5,8,11,26, 29,30,32, 34	3,17,36,40	13	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	"	"	"	"
R19	2F1 "	"	3,14,17,40	35	4,6,9,12, 15,16,18, 23,24,25, 31,37,39, 36	"	"	"	"
R20	2F2 "	"	3,14,36,40	33	4,6,9,12, 15,16,18, 17,23,24, 25,31,37, 39	"	"	"	"
R21	2F3 "	"	3,14,17,36	20	4,6,9,12, 15,16,18, 40,23,24, 25,31,37, 39	"	"	"	"

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TABLE XVIII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R22	2EQ Output crosstalk, delayed, low level	5,8,11,26, 29,30,32, 34	3,14,17,36	19	4,6,9,12, 40,15,16, 18,23,24, 25,31,37, 39	0	500	mV	18,19
R23	1CO "	5,8,11,23, 14,17,25, 26,29,30, 34,36,40	4,6,9,32	28	3,12,15, 16,18,24, 31,37,39	"	"	"	"
R24	2CO "	5,8,11,15, 14,17,39, 26,29,30, 34,36,40	3,18,31,37	38	4,6,9,12, 23,16,24, 25,32	"	"	"	"

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TABLE XIX. Output crosstalk (high-level) test.

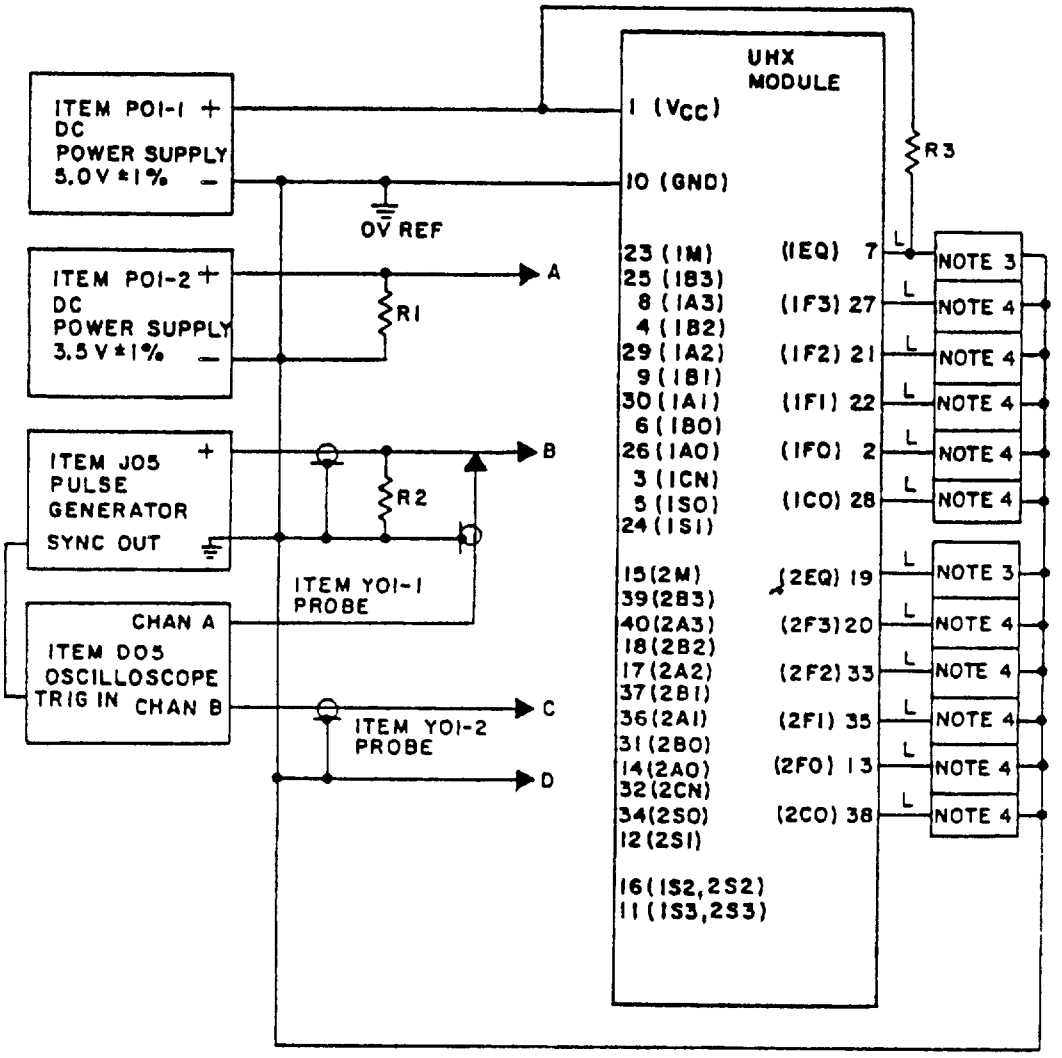
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	2F0 Output crosstalk, high level	5,8,11,14, 15,17,26, 29,30,31, 34,36,40	3,18,37,39	13	4,6,9,12, 16,23,24, 25,32	0	400	mV	18,19
S2	2F1 "	5,8,11,14, 15,17,26, 29,37,30, 34,36,40	3,18,31,39	35	"	"	"	"	"
S3	2F2 "	5,8,11,14, 17,15,26, 29,30,34, 18,36,40	3,31,37,39	33	"	"	"	"	"
S4	2F3 "	5,8,11,14, 15,17,26, 29,30,39, 40,34,36	3,18,31,37	20	"	"	"	"	"
S5	1F0 "	5,8,11,14, 17,23,26, 29,30,34, 36,40,6	4,25,9,32	2	3,12,15, 16,18,24, 31,37,39	"	"	"	"
S6	1F1 "	5,8,11,14, 17,23,26, 29,30,34, 36,40,9	4,6,25,32	22	3,12,15, 16,18,24, 31,37,39	"	"	"	"
S7	1F2 "	5,8,11,14, 17,26,29, 30,23,34, 36,40,4	6,9,25,32	21	3,12,15, 16,18,24, 31,37,39	"	"	"	"
S8	1F3 "	5,8,11,23, 14,17,25, 26,29,30, 34,36,40	4,6,9,32	27	3,12,15, 16,18,24, 31,37,39	"	"	"	"

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TABLE XIX. Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S9	1EQ Output crosstalk, high level	3,5,8,11, 14,17,26, 29,30,34, 36,40	32	7	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	0	400	mV	18,19
S10	2EQ "	5,8,11,14, 17,26,29, 30,32,34, 36,40	3	19	4,6,9,12, 15,16,18, 23,24,25, 31,37,39	"	"	"	"
S11	1CQ "	3,5,11,14, 17,34,36, 40	26,29,30,32	28	4,6,9,12, 15,16,23, 24,25,18, 31,37,39, 8	"	"	"	"
S12	2CQ "	5,8,11,26, 29,30,32, 34	3,14,17,36	38	4,6,9,12, 15,16,18, 23,24,25, 31,37,39, 40	"	"	"	"

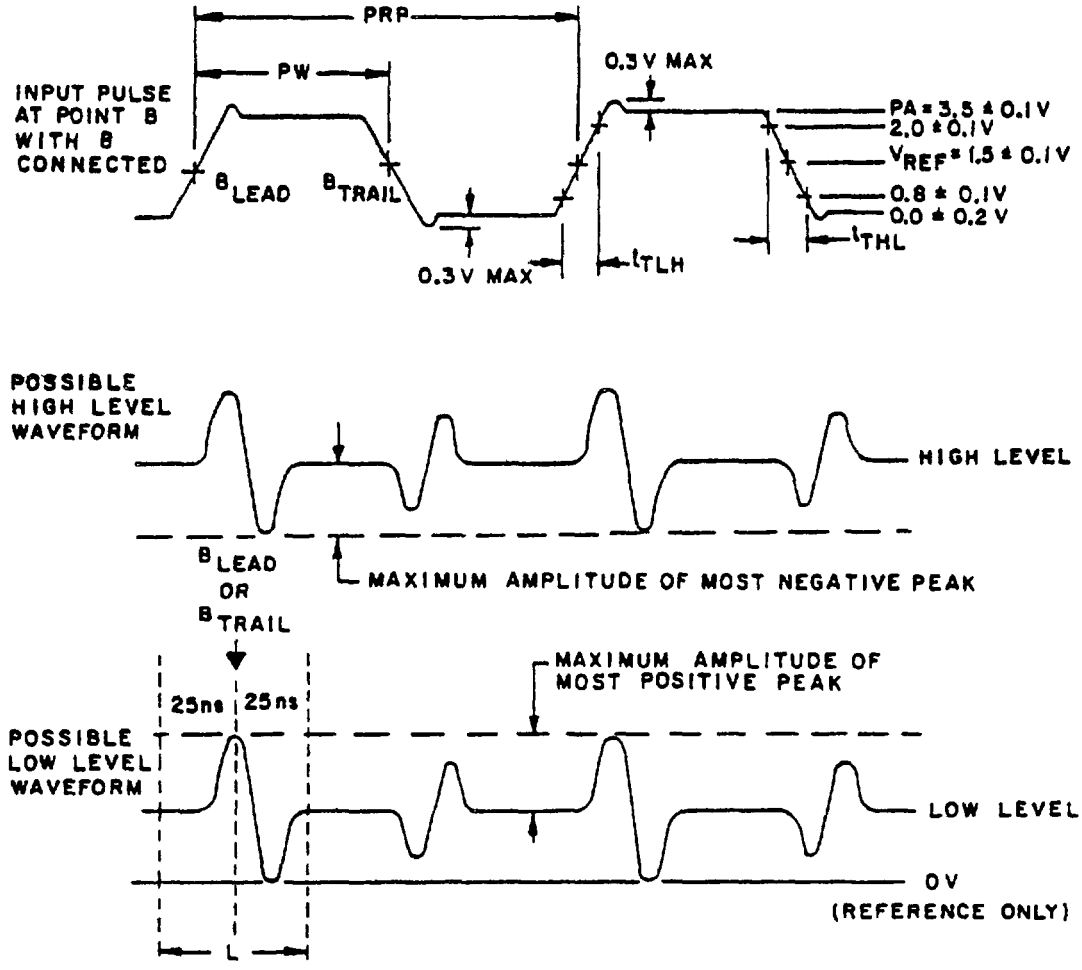
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 1.21 kΩ ±1%, 1/4 W.
3. Use load circuit 2, figure 12, except C_L = 50 pF.
4. Use load circuit 1, figure 12, except C_L = 50 pF.

FIGURE 18. Output crosstalk test circuit for tables XVIII and XIX.

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NOTES
1. Input pulse parameters:
   PW = 200 ±20 ns.
   PRP = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input).
2. For test conditions R13-R24 maximum amplitude of most positive peak must not exceed specified limits outside of time period "L".

FIGURE 19. Output crosstalk test circuit waveforms for tables XVIII and XIX.

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TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	20,21

TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U	Power supply filter capacitance	1	1.6	3.0	µF	22

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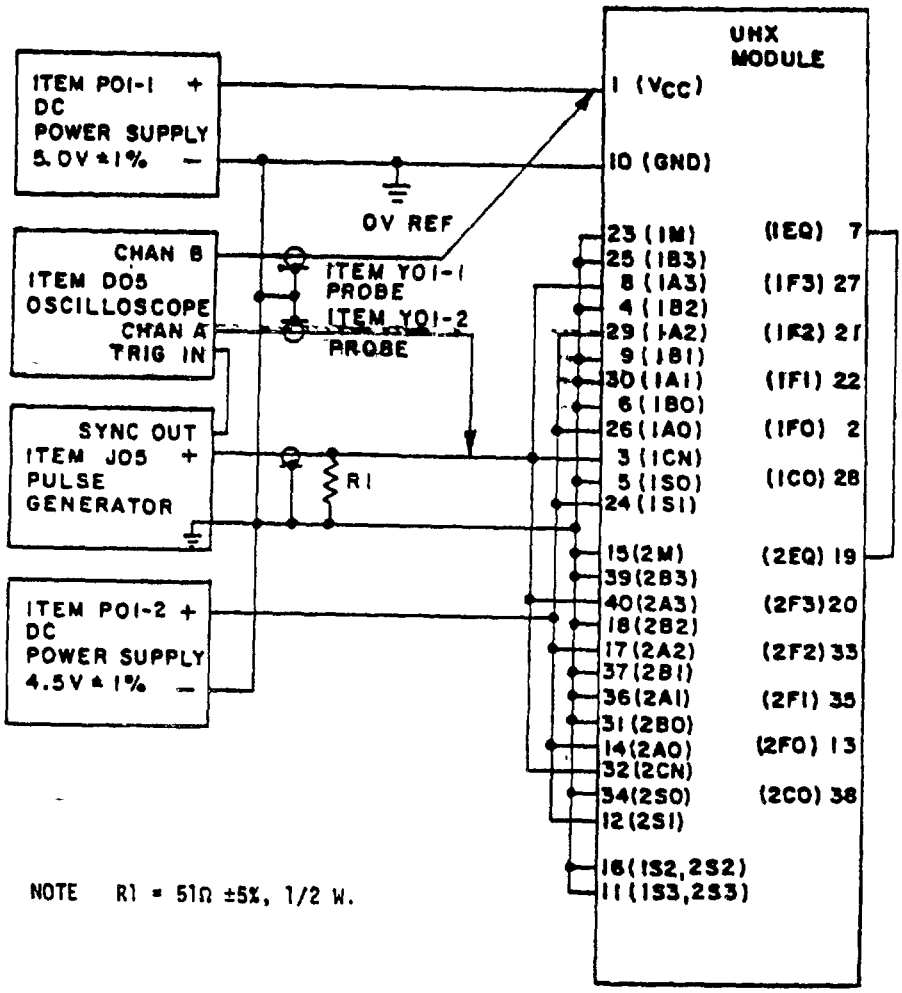
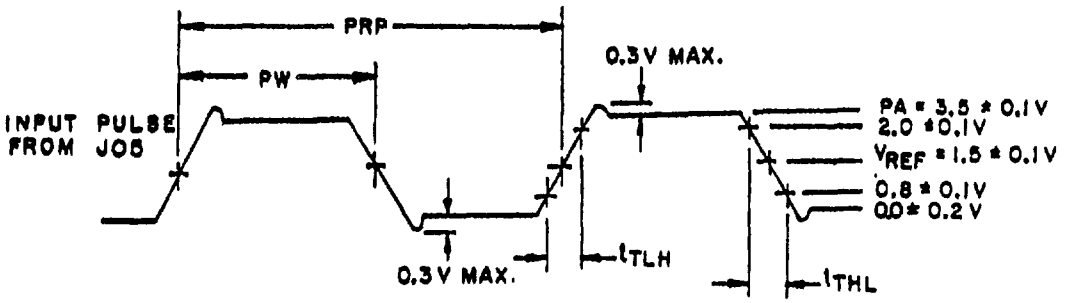
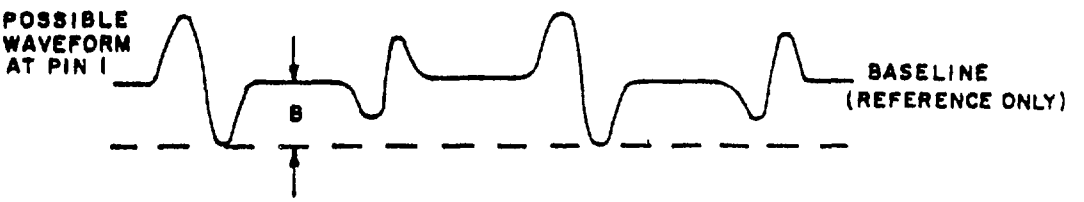


FIGURE 20. Power supply transient amplitude test circuit for table XX.

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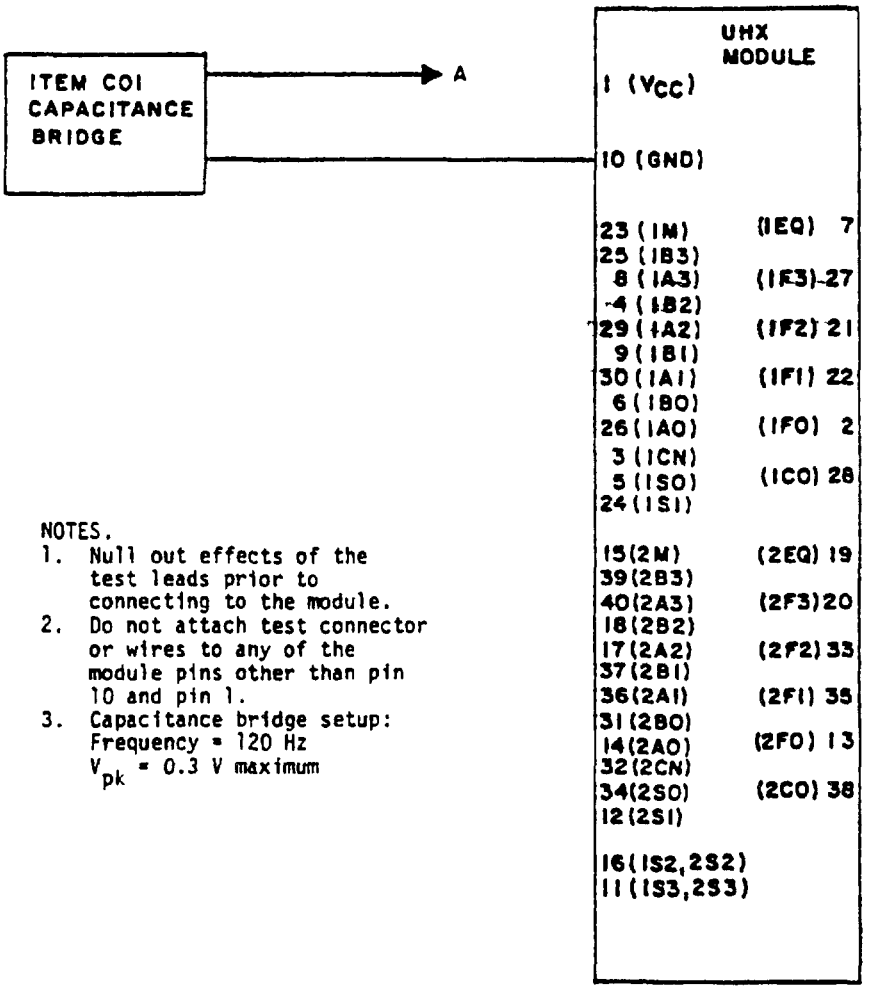




NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at input pins).
2. B = measurement from baseline to most negative excursion of the waveform.

FIGURE 21. Power supply transient amplitude test circuit waveforms for table XX.

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NOTES.
1. Null out effects of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and pin 1.
3. Capacitance bridge setup:
   Frequency = 120 Hz
   V_{pk} = 0.3 V maximum

FIGURE 22. Power supply filter capacitance test circuit for table XXI.

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TABLE XXII. Input capacitance test.

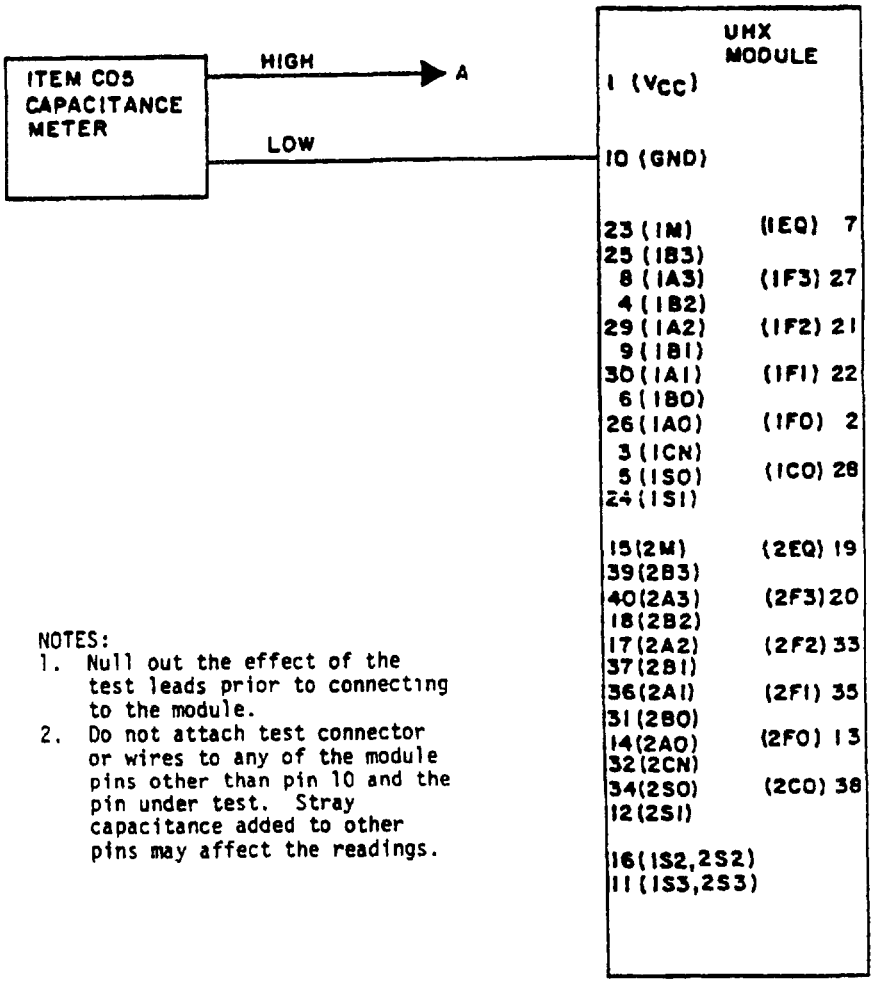
Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
V1	1B3	Input capacitance	25	2	25	pF	23
V2	1A3	"	8	"	"	"	"
V3	1B2	"	4	"	"	"	"
V4	1A2	"	29	"	"	"	"
V5	1B1	"	9	"	"	"	"
V6	1A1	"	30	"	"	"	"
V7	1B0	"	6	"	"	"	"
V8	1A0	"	26	"	"	"	"
V9	1CN	"	3	"	"	"	"
V10	1M	"	23	"	"	"	"
V11	1S0	"	5	"	"	"	"
V12	1S1	"	24	"	"	"	"
V13	2S0	"	34	"	"	"	"
V14	2S1	"	12	"	"	"	"
V15	2M	"	15	"	"	"	"
V16	2CN	"	32	"	"	"	"
V17	2B3	"	39	"	"	"	"
V18	2A3	"	40	"	"	"	"
V19	2B2	"	18	"	"	"	"
V20	2A2	"	17	"	"	"	"
V21	2B1	"	37	"	"	"	"

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TABLE XXII. Input capacitance test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
V22	2A1	Input capacitance	36	2	25	pF	23
V23	2B0	"	31	"	"	"	"
V24	2A0	"	14	"	"	"	"
V25	1S2, 2S2	"	16	"	40	"	"
V26	1S3, 2S3	"	11	"	40	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 23. Input capacitance test circuit for table XXII.

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TABLE XXIII. Functional test.

Test condition	Parameter	Connect			Output logic states												Initial 25°C ambient limits ±5°C		Unit	Test figure
		A to pin	B to pin	C to pin	7 1EQ	2 1FO	22 1F1	21 1F2	27 1F3	28 1CO	19 2EQ	13 2FO	35 2F1	33 2F2	20 2F3	38 2CO	Min	Max		
W1	Functional	3,4,5,6,8,9, 11,18,31,32, 34,37,40	12,14,15,16, 17,23,24,25, 26,29,30,36, 39	2/,3/	H	H	H	H	H	H	H	H	H	H	H	H		1/		24
W2	"	8,11,14,16, 17,26,29,30, 36,40	3,4,5,6,9,12, 15,18,23,24, 25,31,32,34, 37,39	"	H	H	H	H	H	L	H	H	H	H	H	L		"		"
W3	"	3,4,5,11,15, 16,17,18,23, 25,29,32,34, 39	6,8,9,12,14, 24,26,30,31, 36,37,40	"	L	H	H	H	L	L	L	H	H	H	L	L		"		"
W4	"	3,9,11,12,15, 23,24,25,32, 37,39	4,5,6,8,14, 16,17,18,26, 29,30,31,34, 36,40	"	L	L	H	L	H	H	L	L	H	L	H	H		"		"
W5	"	4,8,9,16,17, 18,25,29,30, 36,37,39,40	3,5,6,11,12, 14,15,23,24, 26,31,32,34,	"	H	H	H	H	H	H	H	H	H	H	H	H		"		"
W6	"	6,9,11,15,17, 23,29,31,37	3,4,5,8,12, 14,16,18,24, 25,26,30,32, 34,36,39,40	"	L	H	H	L	H	H	L	H	H	L	H	H		"		"
W7	"	8,9,16,17,29, 30,36,37,40	3,4,5,6,11, 12,14,15,18, 23,24,25,26, 31,32,34,39	"	L	H	H	L	H	L	L	H	H	L	H	L		"		"
W8	"	3,5,8,11,12, 24,25,32,34, 39,40	4,6,9,14,15, 16,17,18,23, 26,29,30,31, 36,37	"	L	H	H	H	L	L	L	H	H	H	L	L		"		"

See footnotes at end of table.

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TABLE XXIII. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states												Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	7 1EQ	2 1F0	22 1F1	21 1F2	27 1F3	28 1C0	19 2EQ	13 2F0	35 2F1	33 2F2	20 2F3	38 2C0	Min	Max		
W9	Functional	4,8,9,11,12, 14,18,24,25, 26,30,36,37, 39,40	3,5,6,15,16, 17,23,29,31, 32,34	2/,3/	L	L	H	H	L	L	L	L	H	H	L	L		1/		24
W10	"	3,4,5,8,11, 12,17,18,24, 25,29,32,34, 39,40	6,9,14,15,16, 23,26,30,31, 36,37	"	L	H	H	L	H	L	L	H	H	L	H	L		"		"
W11	"	6,11,12,16, 17,24,29,30, 31,36	3,4,5,8,9,14, 15,18,23,25, 26,32,34,37, 39,40	"	L	H	L	H	L	L	L	H	L	H	L	L		"		"
W12	"	3,5,6,9,11, 14,17,26,29, 31,32,34,37	4,8,12,15,16, 18,23,24,25, 30,36,39,40,	"	L	L	L	L	H	H	L	L	L	L	H	H		"		"
W13	"	3,4,5,9,11, 15,18,23,25, 30,32,34,36, 37,39	6,8,12,14,16, 17,24,26,29, 31,40	"	L	H	H	L	L	L	L	H	H	L	L	L		"		"
W14	"	8,9,11,25,37, 39,40	3,4,5,6,12, 14,15,16,17, 18,23,24,26, 29,30,31,32, 34,36	"	L	H	L	L	L	L	L	H	L	L	L	L		"		"
W15	"	3,5,6,8,9,14, 16,25,26,30, 31,32,34,36, 37,39,40	4,11,12,15, 17,18,23,24, 29	"	L	H	H	L	H	H	L	H	H	L	H	H		"		"
W16	"	4,11,14,16, 17,18,25,26, 29,39	3,5,6,8,9,12, 15,23,24,30, 31,32,34,36, 37,40	"	L	H	H	L	H	H	L	H	H	L	H	H		"		"

See footnotes at end of table.

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TABLE XXIII Functional test - Continued.

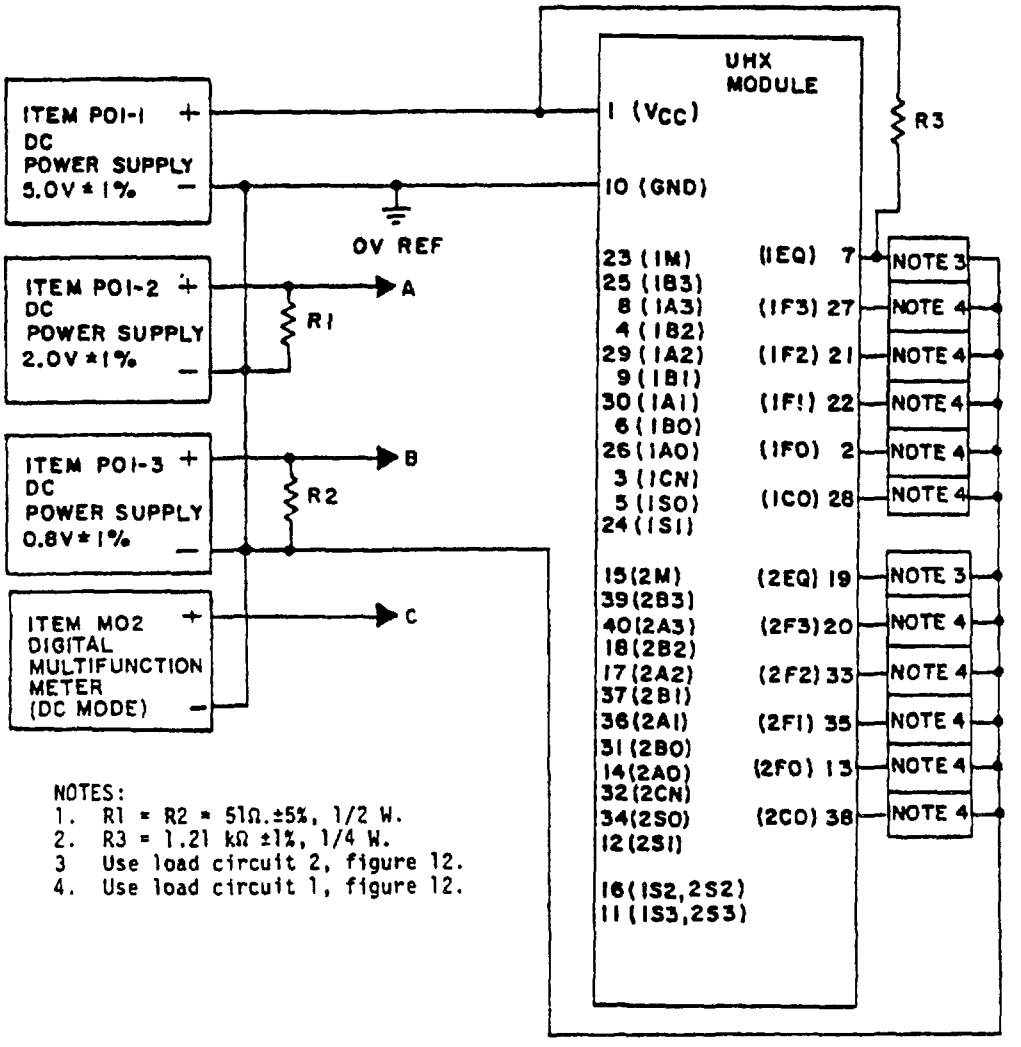
Test condition	Parameter	Connect			Output logic states												Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	7 1EQ	2 1F0	22 1F1	21 1F2	27 1F3	28 1C0	19 2EQ	13 2F0	35 2F1	33 2F2	20 2F3	38 2C0	Min	Max		
W17	Functional	3,4,5,8,9,11, 18,25,30,32, 34,36,37,39, 40	6,12,14,15, 16,17,23,24, 26,29,31	2/,3/	L	L	L	L	H	L	L	L	L	L	H	L		1/		24
W18	"	3,4,5,6,9,11, 12,14,15,18, 23,24,25,26, 31,32,34,37, 39	8,16,17,29, 30,36,40	"	L	H	L	L	L	L	L	H	L	L	L	L		"		"
W19	"	4,6,11,14,15, 18,23,25,26, 30,31,36,39	3,5,8,9,12, 16,17,24,29, 32,34,37,40	"	L	H	L	H	H	H	L	H	L	H	H	H		"		"
W20	"	5,6,9,11,25, 30,31,34,36, 37,39	3,4,8,12,14, 15,16,17,18, 23,24,26,29, 32,40	"	L	L	H	H	H	H	L	L	H	H	H	H		"		"
W21	"	4,5,6,11,18, 25,31,34,39	3,8,9,12,14, 15,16,17,23, 24,26,29,30, 32,36,37,40	"	L	L	H	H	H	H	L	L	H	H	H	H		"		"
W22	"	11,12,16,24	3,4,5,6,8,9, 14,15,17,18, 23,25,26,29, 30,31,32,34, 36,37,39,40	"	L	L	L	L	L	L	L	L	L	L	L	L		"		"
W23	"	5,6,9,12,14, 17,24,26,29, 31,34,37	3,4,8,11,15, 16,18,23,25, 30,32,36,39, 40	"	L	L	L	L	L	L	L	L	L	L	L	L		"		"

1/ Output logic states:
L = low-voltage level = 0.0 to 0.5 V; H = high-voltage level = 2.5 to 4.5 V.

2/ For test conditions W1-W23, verification of the output logic states in footnote 1 is obtained by connecting item M02, point C, to each pin in succession after input conditions have been accomplished.

3/ 1EQ, 1F0-1F3, 1C0, 2EQ, 2F0-2F3, and 2C0 output logic states are measured at output pins 7, 2, 22, 21, 27, 28, 19, 13, 35, 33, 20, and 38, respectively.

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NOTES:
1. R1 = R2 = 51Ω.±5%, 1/2 W.
2. R3 = 1.21 kΩ ±1%, 1/4 W.
3 Use load circuit 2, figure 12.
4. Use load circuit 1, figure 12.

* FIGURE 24. Functional test circuit for table XXIII.

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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